Appendices

The back of the book

look it up, say it, check it

Word Finder — 402 words, A to Z, with meanings — page 2

Symbol Phrasebook — 116 symbols: see it, say it, mean it — page 14

Formula Pages — 361 formulas, Part by Part — page 22

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Everything here was collected automatically while the chapters were being built, so every page reference points to the place where the idea is **properly explained**, with pictures. References look like 24·p3: Chapter 24, page 3 of that chapter's file.

Use the drawers above like a filing cabinet. In Xournal++ you can also tick, circle and scribble on every page here.

Word Finder

Every maths word in the book, A to Z, with a one-line meaning and where it is explained. Numbers like 36·p3 mean Chapter 36, page 3. The thumb tabs on the right edge work like a dictionary's.

A

acute / obtuse / reflex 27·p2
less than 90° / between 90° and 180° / more than 180°

adding vectors (tip to tail) 32·p5
start each vector where the last one ended; the answer runs from the first start to the last end

addition rule 25·p3
p(E or F) = p(E) + p(F) − p(E and F)

algebraic fraction 13·p2
a fraction with letters in the top, bottom or both, e.g. 3/x or (x + 1)/(x − 2)

all solutions 30·p4
trig equations have many answers: find the calculator's one, its partner, then add or subtract whole turns

ambiguous case 31·p8
two sides and a non-included angle can give two different triangles, since sin B = sin(180° − B)

amplitude 29·p5
the height of a wave's peak above its middle line

angle 27·p2
an amount of turning between two directions

angle sum of a triangle 31·p2
the three angles of any triangle add up to 180°

anticlockwise positive 27·p4
maths angles start at the positive x-axis and turn anticlockwise; clockwise is negative

arc length 27·p8
the distance along the curved edge of a slice of circle: s = rθ (θ in radians)

area 5·p5
the amount of flat surface inside a shape, counted in unit squares

area function 36·p8
A(x) = area under f from a fixed start up to x; its derivative is f(x)

area of a triangle (trig) 31·p9
½ × two sides × sine of the angle between them

area under a curve 36·p2
the region between a graph and the x axis; it measures a total (amount, distance, probability)

arithmetic progression (A.P.) 17·p2
a sequence where you add the same number d each time

asymptote 19·p6
a line a graph gets closer and closer to but never reaches, e.g. the axes for y = 1/x

average 24·p2
a single typical value for a data set: mean, median or mode

axes 19·p2
the two number lines of a graph: the x-axis across and the y-axis up (singular: axis)

B

bar chart 23·p3
separate bars (with gaps) whose heights show the frequency of each category

base 9·p2
in a^n, the number a that gets multiplied by itself

base (of a log) 22·p2
the number being raised to a power, written small: the 2 in $\log_2 8$

Bayes' theorem 25·p10
p(a|b) = p(b|a) p(a) / p(b): turns one conditional probability round

bell curve 26·p4
the Normal distribution's graph: symmetric, one central peak, tails fading to zero

Bessel's correction 24·p12
dividing by n − 1 instead of n for a sample, to undo the underestimate of spread

binary 10·p3
base 2: counting with only the digits 0 and 1

bit 10·p3
a binary digit, 0 or 1

BODMAS 1·p10
the order to do operations: Brackets, Orders (powers), Division/Multiplication, Addition/Subtraction

bounds 3·p13
the smallest and largest values a rounded number could really have been

box plot 24·p9
a box from LQ to UQ with a line at the median and whiskers out to the min and max

byte 10·p11
8 bits; can store $2^8 = 256$ different values (0-255)

C

cancel 13·p5
divide the top and bottom of a fraction by the same factor (something multiplying the whole top and the whole bottom)

cardioid 33·p7
the heart-shaped polar curve r = a(1 + cos θ)

CAST rule 29·p3
which trig ratios are positive in each quadrant: All (Q1), Sin (Q2), Tan (Q3), Cos (Q4)

categorical data 23·p2
data that are labels or groups (tea / coffee), not quantities

causation 23·p10
one thing actually makes the other happen

chain rule 35·p4
for a function inside a function: dy/dx = dy/dz × dz/dx (outside rate × inside rate)

chord 34·p3
a straight line joining two points on a curve; its gradient is an average rate

circumference 5·p4
the perimeter of a circle

class width 23·p7
the size of a group in a histogram: upper boundary minus lower boundary

coefficient 8·p4
the number multiplying a letter, e.g. 5 in 5x

collinear 20·p19
lying on the same straight line

common difference 17·p2
d, the fixed amount added each time in an arithmetic progression

common factor 12·p3
something that divides every term; taking it out means writing it once outside a bracket

common log 22·p3
log to base 10, the log button: log 1000 = 3

common ratio 17·p7
r, the fixed multiplier in a geometric progression (any term ÷ the one before)

commutative 1·p7, 16·p6
order doesn't matter: a + b = b + a and a × b = b × a (not true for − or ÷)

compare decimals 3·p3
line up the dots, pad with zeros, compare digits from the left

complement 6·p3
everything in the universal set that is not in the given set

complement rule 25·p2
p(E doesn't happen) = 1 − p(E)

completing the square 15·p13
rewriting $x^2 + bx + c$ as $(x + b/2)^2 - (b/2)^2 + c$; for $a \neq 1$ take a out of the x terms first. Gives vertex form $a(x + p)^2 + q$

complex fraction 13·p12
a fraction with fractions in its top or bottom, e.g. (1 + 1/x)/(1 − 1/x); clear it by multiplying top and bottom by the LCD of the small fractions

composite function 18·p8
one function applied to the output of another, written f(g(x)), fg(x) or (f∘g)(x)

compound angle formula 30·p8
sin(A + B) = sin A cos B + cos A sin B

compound interest 21·p5
interest added each period and then earning interest itself: amount = A × (1 + rate)^t

concentration 4·p17
how much of a substance is in each amount of mixture, e.g. 20% or 60 g per litre

conditional probability 25·p4
p(E | F): the probability of E once you know F happened; F becomes the new sample space

constant 8·p4
a term with no letter: a fixed number

constant of integration (+c) 36·p8
the unknown constant added to every indefinite integral

contingency 7·p7
a statement that is sometimes true and sometimes false in its truth table

contour map 38·p8
a map with lines joining points of equal height; lines close together = steep

contradiction 7·p7
a statement that is false in every row of its truth table, e.g. A ∧ ¬A

contrapositive 7·p9
not Q ⇒ not P; always has the same truth as P ⇒ Q

converge 17·p10
a never-ending sum converges if its total settles towards a fixed number

converse 7·p9
the 'if' statement with its two halves swapped: Q ⇒ P

converting to polar 33·p8
replace x by r cos θ and y by r sin θ, then make r the subject

coordinates 19·p2
the pair (x, y) giving a point's position: x across, then y up

correlation 23·p10
two measurements tend to rise (or fall) together; seen as a sloping cloud on a scatter plot

cosine graph 29·p5
the wave y = cos x: the sine wave shifted π/2 left; starts at its peak, 1

cosine rule 31·p5
$a^2 = b^2 + c^2 - 2bc \cos A$: use it with two sides and the angle between, or with all three sides

cross product 32·p13
a×b: a vector at right angles to a and b, of length |a||b| sin θ

cross-multiply 15·p3
from a/b = c/d go straight to ad = cb; it's multiplying both sides by b and by d

D

data 23·p2
recorded facts: measurements, counts or labels

De Morgan's laws 7·p6
not(A and B) = (not A) or (not B); not(A or B) = (not A) and (not B)

De Morgan's laws (sets) 6·p9
$(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$

decimal places (d.p.) 3·p12
the number of digits after the decimal point

decimal point 3·p2
the dot that separates whole units (left) from parts of a unit (right)

definite integral 36·p13
∫ from a to b: a number, F(b) − F(a), the (signed) area between x = a and x = b

degree 27·p2
1/360 of a full turn, written °

degree-radian conversion 27·p6
× π/180 turns degrees into radians; × 180/π turns radians into degrees

denominator 2·p2
the bottom number of a fraction: how many equal parts make one whole

density 5·p12
mass per unit volume: mass ÷ volume

derivative 34·p4
the gradient function dy/dx: the exact gradient of the curve at every x

derivative of a constant 34·p6
zero: a flat line has no slope

determinant 16·p8
for a 2 × 2 matrix, ad − bc; if it is 0 the matrix has no inverse

deviation 24·p10
how far a value is from the mean: $x - \bar{x}$ (negative if below)

diagonal matrix 16·p7
a square matrix whose only non-zero entries are on the top-left to bottom-right diagonal

diameter 5·p4
distance straight across a circle through the centre; twice the radius

difference of two squares 12·p11
something squared minus something else squared; it always factorises as (a − b)(a + b)

differentiation 34·p4
the process of finding the gradient function (derivative) of a function

digit 1·p2
one of the ten symbols 0 1 2 3 4 5 6 7 8 9

direction angle 32·p8
the angle a vector makes with the positive x-axis: $\tan^{-1}(y/x)$, then the quadrant fix

directional derivative 38·p9
the slope in any direction: $(\partial f/\partial x)$(east part) + $(\partial f/\partial y)$(north part), a dot product with ∇f

discriminant 12·p8
$\Delta = b^2 - 4ac$ for $ax^2 + bx + c$. A perfect square: whole-number brackets. Not a square: square-root brackets. Negative: no real brackets (and no real solutions)

displacement 32·p2
where you ended up compared with the start (how far and which way): a vector

distance 32·p2
how far you travelled in total: a scalar

distance between two points 20·p16
$\sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$: Pythagoras on the run and rise

distribution 26·p2
the pattern of how often each possible value occurs

distributive law 1·p7
a(b + c) = ab + ac: multiply each part of the bracket

doesn't factorise (with whole numbers) 12·p9
a quadratic like $x^2 + 4x + 2$ whose Δ is not a perfect square: no whole-number brackets, only square-root ones. Exam questions that say "factorise" expect whole numbers

domain 18·p4
the set of inputs a function accepts

dominant term 19·p7
the term that decides a graph's shape in some range: the highest power for large x

dot product 32·p9
a·b = |a||b| cos θ = axbx + ayby (+ azbz): a scalar measuring how much two vectors point the same way

dotplot 23·p3
a number line with one dot stacked up for every data value

double angle formulas 33·p10
sin 2A = 2 sin A cos A; cos 2A = cos²A − sin²A = 2cos²A − 1 = 1 − 2sin²A

doubling time 21·p8
the time it takes for a growing quantity to double

dy/dx notation 34·p4
dy/dx is one symbol meaning 'the gradient of y as x changes'; the d's never cancel

E

e 21·p6
the number 2.71828..., the limit of $(1 + 1/n)^n$; the natural base for growth and decay

element (member) 6·p2
one of the things in a set

elimination 15·p5
adding or subtracting whole equations so one unknown cancels out

empty set 6·p10
the set with no elements, written {} or ∅

entry (element) 16·p2
one number inside a matrix; a_{23} is the one in row 2, column 3

equal matrices 16·p4
matrices of the same size whose matching entries are all equal

equals sign (=) Part 0·p4
the two sides have exactly the same value, like a balanced see-saw

equation Part 0·p2, 8·p5
a maths sentence saying two things have the same value (it has an = sign)

equation of a circle 28·p5
$(x − a)^2 + (y − b)^2 = r^2$: centre (a, b), radius r

equivalent fractions 2·p9
fractions with the same value, like 1/2 and 2/4: top and bottom multiplied by the same number

even function 19·p5
f(−x) = f(x): the graph is a mirror image across the y-axis (e.g. x^2, x^4)

event 25·p2
a set of outcomes you are interested in; a subset of the sample space

exact trig values 28·p12
sin, cos, tan of 30°, 45°, 60° written with roots, from the two special triangles

exclusive or (XOR) 7·p5
true when exactly one input is true, not both

explicit / implicit function 35·p12
explicit: y = f(x); implicit: x and y mixed in one equation

exponent 9·p2
another name for the power or index

exponential decay 21·p4
decrease by the same multiplier (between 0 and 1) each step

exponential function 21·p2
a function with the variable in the power, like 2^x or e^x

exponential growth 21·p4
increase by the same multiplier (bigger than 1) each step

expression 8·p4
a combination of numbers, letters and operations with no equals sign

F

factor 1·p13
a number that divides exactly into another: the factors of 12 are 1, 2, 3, 4, 6, 12

factorial 29·p13
n! = n × (n − 1) × ... × 2 × 1; e.g. 4! = 24

factorise 12·p2
rewrite an expression as a product (things multiplied together); the reverse of expanding

factorise by grouping 12·p4
split four terms into two pairs, factorise each pair, then take out the bracket they share

form / message / criticism 23·p12
Lecture 5's three passes for reading a graph: how it's built, what it says, what's wrong or missing

formula Part 0·p9, 8·p5
a shorthand recipe connecting quantities, written with letters

fraction 2·p2
part of a whole, written as top number over bottom number; also means top ÷ bottom

fraction of an amount 2·p8
a/b of N = (N ÷ b) × a

frequency 23·p2
how many times a value (or group of values) occurs

frequency density 23·p7
frequency ÷ class width: the height of a histogram bar, so that area = frequency

frequency table 24·p6
a table listing each value (or class) with how many times it occurs

frozen variable 38·p4
in a partial derivative, every letter except the one you differentiate by acts like a number

function 18·p2
a rule that gives exactly one output for each allowed input

function of two variables 38·p2
one output from two inputs, like h(x, y): a height for each map position

G

geometric progression (G.P.) 17·p7
a sequence where you multiply by the same number r each time

gradient (slope) 20·p2
how steep a line is: how much y goes up for each 1 that x goes across; m in y = mx + c

gradient at a point 34·p12
the value of dy/dx when the point's x-coordinate is substituted in

gradient vector 38·p8
∇f = (∂f/∂x, ∂f/∂y): points steepest uphill, at right angles to the contours

graph of a function 19·p3
all the points (x, f(x)) drawn on axes

Greek letters 8·p12
letters like θ, π, σ used as variables or constants with traditional meanings

grid method (factorising) 12·p8
work out Δ = b² − 4ac and the pair m, n = (b ± √Δ)/2; the grid's edges are then x, n/a on the side and ax, m on top

H

half-life 21·p8
the time it takes for a decaying quantity to halve

HCF (highest common factor) 1·p15
the biggest number that divides exactly into both numbers

hexadecimal (hex) 10·p6
base 16: digits 0-9 then A-F for 10-15

higher derivatives 35·p2
differentiate repeatedly: d^2y/dx^2, d^3y/dx^3, …, d^ny/dx^n

histogram 23·p3
touching bars for numerical data grouped into ranges; area shows frequency

hundredth 3·p2
one of a hundred equal pieces of 1; written 0.01 or 1/100

hypotenuse 28·p2
the longest side of a right-angled triangle, opposite the right angle

I

identity 8·p5
an equation true for every value of the letters, written with ≡

identity matrix 16·p7
the square matrix I with 1s down the diagonal and 0s elsewhere; AI = IA = A

if and only if (iff) 7·p10
P ⇔ Q: P and Q are always both true or both false

image (range) 18·p4
the set of all outputs a function can produce

implication (if... then) 7·p8
P ⇒ Q: false only when P is true but Q is false

implicit differentiation 35·p12
differentiate every term; each y term gets an extra × dy/dx (chain rule); then solve for dy/dx

improper fraction 2·p2
a fraction whose top is bigger than its bottom (more than one whole)

indefinite integral 36·p8
∫ f(x) dx with no limits: a family of functions, + c

independent equations 15·p8
equations that each give genuinely new information; you need as many as there are unknowns

independent events 25·p5
knowing one happened doesn't change the other's probability; then p(E and F) = p(E) × p(F)

inequality 15·p17
a statement that one side is less than (<), greater than (>), at most (≤) or at least (≥) the other; its solution is a range of values

integer 1·p4
a whole number that may be positive, negative or zero: ..., -2, -1, 0, 1, 2, ...

integers 6·p12
whole numbers including 0 and negatives (ℤ)

integral of 1/x 36·p9
ln|x| + c: the one power the reverse power rule can't do

integral sign ∫ 36·p5
a stretched S for 'sum': adds up infinitely many infinitely thin strips

integrand 36·p5
the function being integrated (the strip height)

integration 36·p5
finding the area under a curve; also the reverse of differentiation

integration by parts 37·p5
∫u (dv/dx) dx = uv − ∫v (du/dx) dx: the product rule run backwards

integration by substitution 37·p12
rename an inner chunk as a new letter, swap dx for the new letter's d, integrate, swap back

intercept 19·p4
where a graph crosses an axis: the y-intercept at x = 0, x-intercepts where y = 0

interquartile range 24·p9
UQ − LQ: the spread of the middle half of the data

intersection 6·p3, 20·p17
the elements that are in both sets

interval 6·p11
all the numbers between two end points

inverse function 18·p9
f^{-1}, the function that reverses f: if f(3) = 9 then $f^{-1}(9) = 3$

inverse matrix 16·p8
A^{-1}, the matrix with $A^{-1}A = I$; it undoes multiplying by A

inverse Normal 26·p12
working backwards from an area (a percentage) to the z-score or x-value that gives it

inverse operation 14·p2
the operation that undoes another one: minus undoes plus, divide undoes times

inverse proportion 4·p16
two amounts whose product stays the same: when one doubles, the other halves

inverse trig functions 28·p8
$\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$ (arcsin, arccos, arctan): given the ratio, they give back the angle

irrational number 6·p12
a real number that is not a fraction; its decimals never end or repeat

J

joint probability 25·p3
the probability that two events both happen, p(E ∩ F)

K

kurtosis 26·p13
how peaky or flat a distribution is compared with the Normal (0 for Normal)

L

law of total probability 25·p11
p(B) = p(B|A)p(A) + p(B|A′)p(A′): add up every way B can happen

laws of logs 22·p5
log(ab) = log a + log b, log(a/b) = log a − log b, log a^n = n log a (any base)

LCM (lowest common multiple) 1·p14
the smallest number that is in both times tables

LIATE 37·p7
order for choosing u in parts: Logs, Inverse trig, Algebra, Trig, Exponentials

like terms 11·p2
terms with exactly the same letters raised to the same powers, e.g. $3x^2$ and $-5x^2$

limits of integration 36·p5
the start and end values of x for a definite integral

linear equation 15·p3
an equation where the letter appears only to the power 1 (no x^2, no 1/x)

linearity of integration 36·p12
$\int(f \pm g) = \int f \pm \int g$ and $\int k\,f = k \int f$

local maximum / minimum 34·p15
a hilltop / valley bottom: higher / lower than the points just around it

log scale 22·p4
a scale where equal steps mean multiplying by the same amount (e.g. ×10)

logarithm 22·p2
$\log_b N$ is the power you must raise b to, to get N; it counts how many b's multiply to N

logically equivalent 7·p7
two statements with the same truth value in every row (≡)

lowest common denominator (LCD) 13·p9
the simplest bottom that every fraction's bottom divides into: each different factor of the bottoms, taken the most times it appears in any one bottom

M

magnitude 32·p6
the length of a vector, found by Pythagoras on its components

many-to-one 18·p6
a function where different inputs can give the same output, like x^2 (2 and −2 both give 4)

marginal probability 25·p8
the overall probability of one characteristic, from a margin total

matrix 16·p2
a rectangular table of numbers in rows and columns (plural: matrices)

matrix equation 16·p8
AX = B: a set of simultaneous equations written with one matrix of coefficients A

matrix multiplication 16·p5
entry (i, j) of AB = row i of A paired with column j of B: multiply the pairs and add

mean 24·p2
add all the values and divide by how many there are (the fair-share average)

median 24·p3
the middle value when the data are in order; position (n+1)/2

midpoint 20·p16
the point halfway between two points: average the x-coordinates and the y-coordinates

midpoint (of a class) 24·p6
halfway between a class's boundaries; stands in for its values when estimating a mean

misleading graph 23·p11
a graph whose scale, shape or missing labels give a false impression

mixed number 2·p2
a whole number and a fraction together, such as 1 3/4

mixed partial derivatives 38·p10
$\partial^2 f/\partial x \partial y$: differentiate by one letter then the other; the order doesn't matter

mode 24·p3
the most common value (or busiest group)

modulus (absolute value) 17·p10
the size of a number ignoring its sign, written |x|; $|-0.5| = 0.5$

multiple 1·p13
a number in another number's times table: 12 is a multiple of 4

multiplier 4·p6
the number you multiply by to apply a percentage change, e.g. ×1.1 for +10%

mutually exclusive 25·p3
events that cannot both happen: p(E and F) = 0

N

natural log 22·p3
log to base e (2.718...), written ln: ln e = 1

natural numbers 6·p12
the counting numbers 1, 2, 3, ... (ℕ)

necessary condition 7·p11
Q is necessary for P when P ⇒ Q: P cannot happen without Q

negative area 36·p15
area below the x axis comes out negative in a definite integral

negative correlation 23·p9
one measurement tends to fall as the other rises: the scatter slopes downhill

negative fraction 2·p7
a fraction below zero; -a/b = (-a)/b = a/(-b)

negative of a vector 32·p6
the same arrow pointing the opposite way: every component changes sign

normal (to a curve) 34·p12
the straight line through a point of a curve at right angles to the tangent there

normal distribution 26·p2
the symmetric bell-shaped distribution you get when many small random effects add up

normal probability plot 26·p14
sorted data plotted against the z-scores Normal data would have; straight means Normal

Normal table 26·p6
a table of Φ(z), the area to the left of z under the standard Normal curve

number base 10·p3
how many digits a counting system uses; each column is worth base × the one to its right

numerator 2·p2
the top number of a fraction: how many parts you have

numerical data 23·p2
data that are measured or counted quantities (heights, ages)

numerical gradient 34·p3
$(f(a + h) - f(a)) \div h$ for a small h: an estimate of the gradient at x = a

O

octal 10·p6
base 8: digits 0-7

odd function 19·p5
$f(-x) = -f(x)$: the graph looks the same after a half turn about the origin (e.g. x, x^3)

one-to-one 18·p6
a function where different inputs always give different outputs, like 2x + 3

opposite / adjacent 28·p2
the side across from the angle θ / the side next to θ that isn't the hypotenuse

origin 19·p2
the point (0, 0) where the axes cross

outlier 23·p8
a value far away from the rest of the data

P

parallel lines 20·p3
lines with the same gradient; they never meet

parallel vectors 32·p11
vectors where one is a scalar multiple of the other: b = t a

partial derivative 38·p4
the derivative with respect to one letter, all the others treated as constants

parts formula 37·p5
$\int u\,dv = uv - \int v\,du$

percentage (%) 4·p2
a number of pieces out of 100 equal pieces of the whole

perfect square (expression) 12·p12
an expression that is a bracket times itself, $a^2 \pm 2ab + b^2 = (a \pm b)^2$, e.g. $x^2 + 6x + 9 = (x + 3)^2$

perimeter 5·p4
the total length of the boundary of a shape

period 29·p5
the length of one complete cycle before a graph repeats

perpendicular 20·p9
meeting at a right angle (90°); for lines, gradients multiply to −1

perpendicular bisector 20·p15
the line through the midpoint of AB at right angles to AB

perpendicular vectors 32·p11
vectors at right angles; their dot product is 0

pie chart 23·p3
a circle cut into slices; each slice's angle is its share of 360°

piecewise function 18·p7
a function defined by different formulas on different parts of its domain

place value 1·p2
the position of a digit tells you its size: ones, tens, hundreds...

point of inflection (stationary) 34·p15
a flat 'ledge' where the curve keeps going the same way

polar coordinates 33·p2
(r, θ): distance r from the origin and angle θ anticlockwise from the positive x axis

polar curve 33·p6
the path traced by r = f(θ) as θ turns: the pen is r from the centre at angle θ

polar to Cartesian 33·p3
$x = r\cos\theta$, $y = r\sin\theta$

polar to Cartesian equation 33·p9
swap $r\cos\theta \to x$, $r\sin\theta \to y$, $r^2 \to x^2 + y^2$; multiply by r first if needed

population 24·p12
every member of the group you want to know about

position vector 32·p7
the vector from the origin to a point; for A(x, y) it is xi + yj

positive correlation 23·p9
both measurements tend to rise together: the scatter slopes uphill

posterior probability 25·p14
the updated probability after the evidence, e.g. p(cancer | positive) = 7.8%

power (index, exponent) 8·p4
the small raised number: how many times to multiply by itself

power rule 34·p5
the derivative of a x^n is a $n\,x^{n-1}$: the power comes down in front and drops by one

power series 29·p13
a function written as an endless sum of powers of x, like $\sin x = x - x^3/3! + \ldots$

power set 6·p10
P(S): the set of all subsets of S, including ∅ and S; |P(S)| = 2^n

prefix (metric) 5·p2
a word-start like kilo- or milli- that multiplies a unit by 1000, 1/1000 ...

prime factorisation 1·p14
a number written as primes multiplied together: 60 = 2 x 2 x 3 x 5

prime number 1·p13
a number bigger than 1 whose only factors are 1 and itself

prior probability 25·p14
the probability before new evidence, e.g. p(cancer) = 1%

probability 25·p2
how likely an event is, from 0 (impossible) to 1 (certain); shaded ÷ total when outcomes are equally likely

probability density 26·p5
the height of a smooth distribution curve; areas under it are probabilities

product rule 35·p8
$d(uv)/dx = v\,du/dx + u\,dv/dx$

proper fraction 2·p2
a fraction less than one whole: top smaller than bottom

proper subset 6·p10
a subset that is not the whole set: $A \subset B$ means $A \subseteq B$ and $A \neq B$

proportion 4·p14
two amounts that always stay in the same ratio

Pythagoras in 3D 28·p5
the diagonal of a box: $d^2 = a^2 + b^2 + c^2$

Pythagoras' theorem 28·p3
in a right-angled triangle, hypotenuse² = (one short side)² + (other short side)²

P

Pythagorean identity 30·p2
$\sin^2\theta + \cos^2\theta = 1$: Pythagoras on the unit circle

Q

quadrant 19·p2
one of the four regions the axes divide the plane into

quadrant correction 33·p5
after $\theta = \tan^{-1}(y/x)$: add 180° (π) in quadrants 2 and 3, add 360° (2π) in quadrant 4; points on the axes are read off the sketch

quadratic 12·p8
an expression whose highest power is x^2, like $x^2 + 4x + 3$

quadratic equation 15·p10
an equation with x^2 as the highest power, e.g. $x^2 + 6x + 5 = 0$

quadratic formula 15·p12
$x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$, the solutions of $ax^2 + bx + c = 0$; the roots are Chapter 12's pair with the sign flipped, divided by a

quantifier 7·p12
'for all' (∀) or 'there exists' (∃): says how many things a statement covers

quartiles 24·p9
the values a quarter (LQ) and three quarters (UQ) of the way through the ordered data

quotient rule 35·p10
$d(u/v)/dx = (v\,du/dx - u\,dv/dx) \div v^2$

R

r sin(θ + α) form 30·p8
$a \sin\theta + b \cos\theta$ rewritten as one wave $r \sin(\theta + \alpha)$, $r = \sqrt{(a^2 + b^2)}$, $\tan\alpha = b/a$

radian 27·p5
the angle made at the centre by an arc as long as the radius; 2π radians = 360°

radius 5·p4
distance from the centre of a circle to its edge

range 24·p9
largest value minus smallest value

rate of change 34·p10
how fast one quantity changes as another does: the derivative

ratio 4·p9
how many parts of one thing for every so many parts of another, written a : b

rational number 6·p12
a number that can be written as a fraction of two integers (ℚ)

real numbers 6·p12
all numbers on the number line (ℝ)

reciprocal 2·p8
a fraction flipped upside down: the reciprocal of 4/5 is 5/4, of 4 (= 4/1) is 1/4

recurring decimal 3·p10
a decimal whose digits repeat forever, e.g. 0.333...; shown with dots

reference angle 29·p3
the acute angle between the radius and the x-axis; trig of θ = ± trig of it

remainder 1·p9
what is left over when a number doesn't share out exactly

restricted domain 18·p5
a domain deliberately limited, e.g. $x \geq 0$, often to make a function one-to-one

reverse chain rule 37·p16
$\int f(g(x))\,g'(x)\,dx = F(g(x)) + C$: substitution undoes the chain rule

reverse power rule 36·p6
$\int a\,x^n\,dx = a\,x^{n+1} \div (n + 1) + c$, for any n except −1

right angle 27·p2
a quarter turn, 90°, marked with a little square

right-hand rule 32·p13
first finger along a, second finger along b, thumb gives the direction of a×b

rounding 3·p12
replacing a number with the nearest one on a coarser ruler

S

sample 24·p12
the members you actually measured, used to estimate the population

sample space 25·p2
the set S of every possible outcome

scalar 16·p4, 32·p2
an ordinary single number (as opposed to a matrix or vector)

scalar multiple 32·p6
k times a vector: k times as long, same direction (reversed if k is negative)

scalene / isosceles / equilateral 31·p2
no equal sides / two equal sides / all three sides equal

scatter plot 23·p3
one dot per item, placed by two measurements (x, y), to see whether they are related

second derivative 34·p17
d^2y/dx^2: the derivative of the derivative; positive at a minimum, negative at a maximum

sector area 27·p8
the area of a slice of circle: $A = \frac{1}{2}r^2\theta$ (θ in radians)

sequence 17·p2
an ordered list of numbers following a rule, e.g. 2, 6, 10, 14, ...

series 17·p2
the sum of the terms of a sequence, e.g. 2 + 6 + 10 + 14

set 6·p2
a collection of distinct things, written inside curly brackets { }

set-builder notation 6·p11
{x | rule}: the set of all x that obey the rule

sigma notation 17·p6
Σ with a start value below and a stop value above: add up the expression for every counter value

significant figures (s.f.) 3·p16
digits counted from the first non-zero digit; how precise a number is

similar triangles 28·p6
triangles with the same angles: one is a zoomed copy of the other, so side ratios match

simplify a fraction 2·p10
divide top and bottom by the same number (ideally the HCF) so the numbers are as small as possible

simultaneous equations 15·p5
two (or more) equations about the same unknowns, which must all be true at once

sine graph 29·p4
the wave y = sin x: starts at 0, peaks at 1 (90°), back to 0 (180°), down to −1 (270°), 0 at 360°

sine rule 31·p3
a/sin A = b/sin B = c/sin C: use it when you know a side and its opposite angle

sine, cosine, tangent 28·p6
the ratios opp/hyp, adj/hyp and opp/adj of a right-angled triangle, fixed by the angle

singular matrix 16·p8
a square matrix with determinant 0; it has no inverse

size of a matrix 16·p2
(number of rows) × (number of columns), said "2 by 3"

skewed data 24·p7
data with a long tail on one side; the mean is dragged towards the tail

small changes formula 38·p11
δz ≈ (∂z/∂x)δx + (∂z/∂y)δy: add up each input's effect

SOHCAHTOA 28·p6
memory aid: Sin = Opp/Hyp, Cos = Adj/Hyp, Tan = Opp/Adj

solution 15·p2
a value of the letter that makes the equation true (also called a root)

solve 15·p2
find the value(s) of the letter that make an equation true

solving exponential equations 22·p7
take logs of both sides and use log a^n = n log a to bring the unknown down

speed 5·p12
distance travelled per unit of time: distance ÷ time

split the middle 12·p6
for $ax^2 + bx + c$: find two numbers that multiply to ac and add to b, use them to split bx into two terms, then factorise in pairs

square root Part 0·p3, 9·p3
the number which, multiplied by itself, gives the number you started with

squared Part 0·p3
multiplied by itself: 5 squared is 5 x 5

standard derivatives 34·p9
the basic list: constants, powers, ln x, e^(kx), sin, cos, tan

standard deviation 24·p11
the square root of the variance; a typical distance from the mean, in the data's units

standard form 9·p15
a number written as a × 10^n with 1 ≤ a < 10

standard integrals 36·p12
the basic list: k, powers, 1/x, e^(kx), sin kx, cos kx, tan x

standard Normal distribution 26·p4
the Normal distribution with mean 0 and standard deviation 1

statement 7·p2
a sentence that is either true or false

stationary point 34·p15
a point where the gradient is zero (the tangent is flat)

stem and leaf diagram 23·p7
each value split into stem (leading digit) and leaf (last digit); leaves listed beside their stem

strip width h 36·p3
h = (b − a) ÷ n: the width of each strip

subject of a formula 14·p2
the letter on its own on one side of the = sign

subscript 16·p2
a small number or letter written low after a symbol, used as a label: a_1, x_2, a_{23}

subset 6·p10
a set whose elements are all inside another set

substitute Part 0·p7
replace a letter with a number and work out the value

S

substitution 8·p6
replacing letters by given numbers, then working out the value

substitution limits 37·p16
when substituting in a definite integral, convert the limits into the new letter

substitution method 15·p7
rearrange one equation for one letter and put it into the other equation

sufficient condition 7·p11
P is sufficient for Q when P ⇒ Q: P on its own guarantees Q

sum of squares 24·p11
add up the squared deviations $\Sigma(x - \bar{x})^2$

sum rule (differentiation) 34·p6
differentiate a sum or difference term by term

sum-product rule 12·p6
to factorise $x^2 + bx + c$, find two numbers that add to b and multiply to c; they go in the brackets $(x + p)(x + q)$

surface area 5·p10
the total area of all the faces of a solid

T

tabular integration 37·p8
D/I table for repeated parts: differentiate u down to 0, integrate the other, alternate signs

tangent (to a curve) 34·p2
the straight line that just touches a curve at one point, going the same way as the curve there

tangent equation 34·p14
$y - y_1 = f'(a)(x - a)$: the line through (a, f(a)) with the curve's gradient there

tangent graph 29·p6
y = tan x: repeats every π, with vertical asymptotes where cos x = 0 (90°, 270°, ...)

tautology 7·p7
a statement that is true in every row of its truth table, e.g. A ∨ ¬A

tenth 3·p2
one of ten equal pieces of 1; written 0.1 or 1/10

term 8·p4, 17·p2
one piece of an expression, separated from the others by + or − signs

time constant 21·p7
for e^(−t/τ), the time τ in which the quantity drops to 1/e ≈ 37% of its value

transformation 19·p9
a change to a formula that moves, stretches or flips its graph

transpose 14·p1
rearrange a formula so that a different letter is on its own

trapezium rule 36·p3
estimate an area with n sloping-topped strips: h/2 × (first + 2 × middles + last)

tree diagram 25·p11
branches for each outcome of each stage; multiply along a path, add between paths

triangle notation 31·p2
angles A, B, C at the corners; sides a, b, c opposite them

trig equation 30·p4
a statement like sin θ = 0.5 that is true only for particular angles

trig identity 30·p2
a trig statement that is true for every angle, e.g. $\sin^2\theta + \cos^2\theta = 1$

trig of any angle 29·p2
cos θ and sin θ are the x- and y-coordinates of the point at angle θ on the unit circle

trig transformations 29·p8
y = A sin(Bx) + D: A stretches the height, B squashes the period to 2π/B, D lifts it

truncate 3·p15
chop off digits after a point without rounding

truth table 7·p3
a table listing every combination of T/F inputs and the result

truth value 7·p2
whether a statement is true (T, 1) or false (F, 0)

turning point 34·p15
where a curve changes from going up to going down, or down to up

two-way table 25·p8
a table counting every combination of two characteristics, with totals in the margins

U

undefined 13·p2
has no value; a fraction is undefined when its bottom equals 0

uniform distribution 26·p2
every value equally likely: a flat histogram

union 6·p3
all elements in either set (or both), listed once

unit 5·p2
the fixed amount you count with: a metre, a gram, a second...

unit circle 29·p2
the circle of radius 1 centred at the origin; the point at angle θ is (cos θ, sin θ)

unit vector 32·p3
a vector of length exactly 1; i, j, k point along the x, y, z axes

unit vector of a 32·p7
â = a ÷ |a|: the vector of length 1 pointing the same way as a

unitary method 4·p14
scale a known pair down to 1 unit, then up to the amount you want

universal set 6·p3
the set of everything under discussion, written ξ or U

V

variable Part 0·p6, 8·p2
a letter standing for a number that can change

variance 24·p11
the mean of the squared deviations; σ² for a population

vector 32·p2
a quantity with size AND direction (velocity, displacement, force)

vector notation 32·p3
an arrow over the letter, bold type, or an underline all mark a vector

Venn diagram 6·p4
overlapping circles inside a rectangle, showing how sets share elements

vertex 19·p9
the turning point of a parabola: (h, k) for y = a(x − h)² + k

vertex form 15·p13
a quadratic written a(x + p)² + q; its vertex (turning point) is (−p, q)

vertical line test 18·p6
a graph is a function if no vertical line crosses it more than once

volume 5·p9
the amount of 3-D space inside a solid, counted in unit cubes

Y

y-intercept 20·p2
c in y = mx + c: the value of y where the line crosses the y-axis (x = 0)

Z

z-score 26·p6
how many standard deviations a value is above (+) or below (−) the mean: z = (x − μ)/σ

zero product rule 15·p10
if A × B = 0 then A = 0 or B = 0

#

3 × 3 determinant 32·p16
cover each top-row entry's row and column, take the 2 × 2 determinant left over, and combine with signs + − +

Symbol Phrasebook

See it, say it, mean it. The middle column is how you'd read the symbol out loud; saying it is half of understanding it (Chapter 0).

Start here: The language of maths

π	*pie*	a fixed number, about 3.14159, from circles **Part 0·p10**
$=$	*equals*	has the same value as **Part 0·p4**
$\neq$	*is not equal to*	different value **Part 0·p4**
$<$	*is less than*	smaller than **Part 0·p4**
$>$	*is greater than*	bigger than **Part 0·p4**
$\leq$	*less than or equal to*	smaller or the same **Part 0·p4**
$\geq$	*greater than or equal to*	bigger or the same **Part 0·p4**
$\approx$	*approximately equal to*	roughly the same **Part 0·p4**
$\equiv$	*identically equal to*	true for every possible number **Part 0·p5**
$\gg$	*much greater than*	far bigger **Part 0·p5**
$\sqrt{}$	*square root of*	which number times itself gives this **Part 0·p3**
x^2	*x squared*	x times x **Part 0·p3**
x_1	*x one*	a name tag: the first x **Part 0·p10**

Part 1: Numbers

$0.\dot{3}$	*nought point three recurring*	the 3 repeats forever **3·p10**
$\approx$	*is approximately equal to*	close to, but not exactly **3·p10**

$\%$	*per cent*	out of 100, so 25% means 25/100 **4·p2**
$a : b$	*a to b*	a parts of one thing for every b parts of the other **4·p9**
$::$	*is in the same ratio as*	a:b :: c:d means a/b = c/d **4·p10**
cm^2	*square centimetres*	a count of 1 cm x 1 cm squares **5·p5**
cm^3	*cubic centimetres*	a count of 1 cm cubes **5·p9**
$\mathrm{m\,s}^{-1}$	*metres per second*	same as m/s **5·p12**
$\{\ \}$	*the set of / braces*	curly brackets that hold the members of a set **6·p2**
$\in$	*is an element of / is in*	x ∈ A means x belongs to set A **6·p2**
$\notin$	*is not an element of*	x ∉ A means x is not in A **6·p2**
$n(A)$	*n of A / the size of A*	how many elements are in set A **6·p2**
ξ	*ksee (or 'the universal set')*	everything being talked about **6·p3**
$\cap$	*intersection / and*	elements in both sets **6·p3**
$\cup$	*union / or*	elements in either set (or both) **6·p3**
A'	*A complement / not A*	everything in ξ that is not in A **6·p3**
$\subset$	*is a subset of*	every element of the left set is in the right set **6·p10**
$\varnothing$	*the empty set*	a set with nothing in it **6·p10**
$\subseteq$	*is a subset of (or equal to)*	every element of the left set is in the right set **6·p10**
$\mid$	*such that*	separates the variable from the rule it must obey **6·p11**

$[a, b]$	the closed interval from a to b	all numbers from a to b, ends included **6·p11**
(a, b)	the open interval from a to b	all numbers between a and b, ends excluded **6·p11**
$[a, \infty)$	a to infinity	all real numbers greater than or equal to a **6·p11**
$\mathbb{N}$	N, the natural numbers	counting numbers 1, 2, 3, ... **6·p12**
$\mathbb{Z}$	Z, the integers	whole numbers including negatives and 0 **6·p12**
$\mathbb{Q}$	Q, the rational numbers	anything that can be written as a fraction **6·p12**
$\mathbb{R}$	R, the real numbers	every point on the number line **6·p12**
$\neg P$	not P	true exactly when P is false **7·p2**
$A \wedge B$	A and B	true only when both A and B are true **7·p3**
$A \vee B$	A or B	true when at least one of A, B is true **7·p4**
$A \oplus B$	A x-or B (exclusive or)	true when exactly one of A, B is true **7·p5**
$\equiv$	is logically equivalent to	same truth value in every row of the table **7·p7**
$P \Rightarrow Q$	P implies Q / if P then Q	false only when P is true and Q is false **7·p8**
$P \Leftrightarrow Q$	P if and only if Q (iff)	P and Q are both true or both false **7·p10**
$\forall$	for all	the statement holds for every value **7·p12**
$\exists$	there exists	at least one value makes the statement true **7·p12**

Part 2: Algebra

$\equiv$	*is identically equal to*	true for all values **8·p5**
α	*alpha (AL-fa)*	angles; radiation (α particles) **8·p12**
β	*beta (BEE-ta)*	angles; radiation **8·p12**
γ	*gamma (GAM-a)*	angles; gamma rays **8·p12**
$\delta,\ \Delta$	*delta (DEL-ta)*	a small change (Δx: change in x) (Ch. 20, 34) **8·p12**
ε	*epsilon (EP-si-lon)*	a tiny amount; errors **8·p12**
θ	*theta (THEE-ta)*	the most common angle letter (Ch. 27-31) **8·p12**
ϕ	*phi (FYE (or fee))*	another angle **8·p12**
λ	*lambda (LAM-da)*	wavelength; eigenvalues **8·p12**
μ	*mu (MEW)*	the mean in statistics (Ch. 24, 26) **8·p12**
$\sigma,\ \Sigma$	*sigma (SIG-ma)*	σ: spread (Ch. 24); Σ: add up (Ch. 17) **8·p12**
π	*pi (PIE)*	3.14159..., the circle number **8·p12**
ω	*omega (oh-MEE-ga)*	angular speed; spinning **8·p12**
χ	*chi (KYE)*	the chi-squared test in statistics **8·p12**
a^n	*a to the power n*	n copies of a multiplied together **9·p2**
$\sqrt{x}$	*the square root of x*	the positive number whose square is x **9·p3**
$\sqrt[n]{x}$	*the n-th root of x*	the number whose n-th power is x **9·p3**
$\pm$	*plus or minus*	both the + and − versions **9·p3**

10110_2	*one-zero-one-one-zero in base two*	the subscript gives the base **10·p4**
$\pm$	*plus or minus*	both the + version and the − version are answers **15·p10**
a_{ij}	*a sub i j*	the entry in row i, column j of matrix A **16·p2**
A^{-1}	*A inverse*	the matrix that undoes A: $A^{-1}A = I$ **16·p8**
u_n	*u sub n*	the nth term of a sequence **17·p2**
S_n	*S sub n*	the sum of the first n terms **17·p4**
$\sum$	*sigma (sum)*	add up the expression for each value of the counter from the bottom number to the top **17·p6**
S_∞	*S infinity*	the sum of all the terms of a never-ending series **17·p10**
$\lvert r \rvert$	*mod r (the modulus of r)*	the size of r ignoring its sign: $\lvert -3 \rvert = 3$ **17·p10**

Part 3: Functions and graphs

$\mapsto$	*maps to*	the input on the left is turned into the output on the right **18·p2**
$f(x)$	*f of x*	the output of function f when the input is x **18·p2**
$f \circ g$	*f after g (f compose g)*	the composite function f(g(x)): g first, then f **18·p8**
f^{-1}	*f inverse*	the function that undoes f **18·p9**
e	*e (Euler's number)*	2.71828..., the base of natural growth **21·p6**
$\to \infty$	*tends to infinity*	gets bigger and bigger without end **21·p6**
$\exp(x)$	*exp of x*	another way to write e^x **21·p6**

$\log_b N$	*log to the base b of N*	the power of b that gives N **22·p2**
$\ln$	*L-N, natural log*	log to base e **22·p3**

Part 4: Data and chance

$\bar{x}$	*x bar*	the mean of a sample of x-values **24·p2**
σ	*sigma (lower case)*	standard deviation of a population **24·p11**
μ	*mew*	the mean of a whole population **24·p12**
$p(E)$	*p of E*	the probability that event E happens **25·p2**
$p(E \mid F)$	*p of E given F*	probability of E, knowing that F has happened **25·p4**
$\Phi(z)$	*fie of z (capital phi)*	area under the standard Normal curve to the left of z **26·p6**

Part 5: Geometry and trigonometry

$\circ$	*degrees*	360° is one full turn **27·p2**
θ	*theta*	a common letter for an unknown angle **28·p2**
$\sin\theta$	*sine theta (say 'sign')*	opp ÷ hyp for the angle θ **28·p6**
$\sin^{-1}$	*inverse sine, or arcsine*	the angle whose sine is ... **28·p8**
$n!$	*n factorial*	n × (n−1) × ... × 1 **29·p13**
$\sin^2\theta$	*sine squared theta*	$(\sin\theta)^2$, the sine worked out first, then squared **30·p2**
$\vec{a}$	*vector a*	a vector named a (also bold a, or a underlined) **32·p3**
$\overrightarrow{AB}$	*vector A B*	the vector that takes you from point A to point B **32·p3**

$\mathbf{i},\ \mathbf{j},\ \mathbf{k}$	*i, j, k*	unit vectors along x, y and z (also written with hats) **32·p3**
$\lvert\mathbf{a}\rvert$	*the magnitude of a*	the length of vector a **32·p6**
$\hat{\mathbf{a}}$	*a hat*	the unit vector (length 1) in the direction of a **32·p7**
$\mathbf{a} \cdot \mathbf{b}$	*a dot b*	the dot (scalar) product: a plain number **32·p9**
$\mathbf{a} \times \mathbf{b}$	*a cross b*	the cross (vector) product **32·p13**
(r, θ)	*r, theta*	a point in polar coordinates: radius then angle **33·p2**

Part 6: Calculus

h	*h*	a small step along the x axis **34·p3**
$\dfrac{dy}{dx}$	*dee y by dee x*	the derivative: gradient of y with respect to x **34·p4**
$f'(x)$	*f dash of x*	the derivative of the function f **34·p4**
δx	*delta x*	a small change in x **34·p4**
$\dot{y}$	*y dot*	Newton's notation for the derivative of y with respect to time, dy/dt **34·p4**
$\sec x$	*seck x (secant)*	1 ÷ cos x **34·p9**
$\dfrac{d^2y}{dx^2}$	*dee two y by dee x squared*	the second derivative **34·p17**
$\displaystyle\int_a^b f(x)\,dx$	*the integral from a to b of f of x, dee x*	the exact area under y = f(x) between x = a and x = b **36·p5**
$\dfrac{\partial f}{\partial x}$	*partial f by partial x (or del f by del x)*	the slope of f when only x changes; every other letter is frozen as a constant **38·p4**
$f_x,\ f_y$	*f sub x, f sub y*	short for ∂f/∂x and ∂f/∂y **38·p4**

∇f	*grad f (or del f)*	the vector of partial derivatives; points the steepest way uphill **38·p8**

Formula Pages

Every boxed formula from the book in one place, Part by Part. The label says what it's for; the reference says where it's explained and practised. The standard derivative and integral tables come last.

Part 1: Numbers

Adding a negative
Whole numbers · 1·p18
$$a + (-b) = a - b$$

Subtracting a negative
Whole numbers · 1·p18
$$a - (-b) = a + b$$

× and ÷ signs
Whole numbers · 1·p18
$$(-a) \times (-b) = ab, \quad (-a) \times b = -ab$$

Fraction bar
Whole numbers · 1·p18
$$\frac{a+b}{c+d} = (a+b) \div (c+d)$$

Powers of negatives
Whole numbers · 1·p18
$$(-3)^2 = 9, \qquad -3^2 = -9$$

Commutative
Whole numbers · 1·p18
$$a + b = b + a, \quad a \times b = b \times a$$

Distributive law
Whole numbers · 1·p18
$$a(b+c) = ab + ac$$

Division
Whole numbers · 1·p18
$$a \times b = c \iff c \div b = a$$

HCF and LCM
Whole numbers · 1·p18
$$\mathrm{HCF}(a,b) \times \mathrm{LCM}(a,b) = a \times b$$

Fraction = division
Fractions · 2·p15
$$\frac{a}{b} = a \div b, \quad n = \frac{n}{1}$$

Equivalent fractions
Fractions · 2·p15
$$\frac{a}{b} = \frac{a \times k}{b \times k} = \frac{a \div k}{b \div k}$$

Mixed to improper
Fractions · 2·p15
$$p\tfrac{a}{b} = \frac{pb + a}{b}$$

Add / subtract
Fractions · 2·p15
$$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$$

Multiply
Fractions · 2·p15
$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

Divide
Fractions · 2·p15

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

Reciprocal
Fractions · 2·p15

$$\frac{a}{b} \to \frac{b}{a}, \qquad \frac{a}{b} \times \frac{b}{a} = 1$$

Compare
Fractions · 2·p15

$$\frac{a}{b} > \frac{c}{d} \iff ad > bc$$

Negative fractions
Fractions · 2·p15

$$-\frac{a}{b} = \frac{-a}{b} = \frac{a}{-b}$$

Fraction of an amount
Fractions · 2·p15

$$\frac{a}{b} \text{ of } N = \frac{N}{b} \times a$$

Decimal to fraction
Decimals · 3·p19

$$0.abc = \frac{abc}{1000}$$

Fraction to decimal
Decimals · 3·p19

$$\frac{a}{b} = a \div b$$

Divide
Decimals · 3·p19

$$\frac{a}{b} = \frac{10a}{10b}$$

Recurring to fraction
Decimals · 3·p19

$$x = 0.\dot{2}\dot{7} : \;\; 100x - x = 27 \Rightarrow x = \frac{27}{99}$$

Bounds
Decimals · 3·p19

$$\text{value} - \tfrac{1}{2}\text{step} \leq x < \text{value} + \tfrac{1}{2}\text{step}$$

Bounds of a ÷
Decimals · 3·p19

$$\max \frac{a}{b} = \frac{a_{\max}}{b_{\min}}, \qquad \min \frac{a}{b} = \frac{a_{\min}}{b_{\max}}$$

percentage of an amount
Percentage and ratio · 4·p4

$$x\% \text{ of } y = \frac{x}{100} \times y$$

percentage increase / decrease
Percentage and ratio · 4·p6

$$\text{new} = \text{old} \times \frac{100 \pm r}{100}$$

percentage change
Percentage and ratio · 4·p7

$$\frac{\text{new} - \text{old}}{\text{old}} \times 100$$

reverse percentage
Percentage and ratio · 4·p8

$$\text{original} = \frac{\text{new}}{1 \pm \frac{r}{100}}$$

cross-multiplication
Percentage and ratio · 4·p15

$$\frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$$

Per cent
Percentage and ratio · 4·p19

$$x\% = \frac{x}{100}$$

Whole from a part
Percentage and ratio · 4·p19

$$\text{whole} = \text{part} \div \frac{p}{100}$$

Increase / decrease by r%
Percentage and ratio · 4·p19

$$\text{new} = \text{old} \times \left(1 \pm \frac{r}{100}\right)$$

Ratio as fractions
Percentage and ratio · 4·p19

$$a : b \Rightarrow \frac{a}{a+b} \text{ and } \frac{b}{a+b} \text{ of the total}$$

Ratio to percentage
Percentage and ratio · 4·p19

$$\frac{\text{part}}{\text{sum of parts}} \times 100$$

Inverse proportion
Percentage and ratio · 4·p19

$$x \times y = \text{constant}$$

Mixing
Percentage and ratio · 4·p19

$$\frac{\text{total A}}{\text{total volume}} \times 100$$

perimeter of a rectangle
Measurement · 5·p4

$$P = 2(l + w)$$

circumference of a circle
Measurement · 5·p4

$$C = \pi d = 2\pi r$$

area of a rectangle
Measurement · 5·p5

$$A = lw$$

area of a triangle
Measurement · 5·p5

$$A = \tfrac{1}{2}bh$$

area of a circle
Measurement · 5·p5

$$A = \pi r^2$$

area of a trapezium
Measurement · 5·p6

$$A = \tfrac{1}{2}(a + b)h$$

volume of a cuboid
Measurement · 5·p9

$$V = lwh$$

volume of a cylinder
Measurement · 5·p9

$$V = \pi r^2 h$$

surface area of a cuboid
Measurement · 5·p10

$$S = 2(lw + lh + wh)$$

speed
Measurement · 5·p12

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

Rectangle, triangle
Measurement · 5·p15

$$A = lw, \qquad A = \tfrac{1}{2}bh$$

Parallelogram, trapezium
Measurement · 5·p15

$$A = bh, \qquad A = \tfrac{1}{2}(a+b)h$$

Volume
Measurement · 5·p15

$$V = lwh, \qquad V = \pi r^2 h$$

Speed
Measurement · 5·p15

$$s = \frac{d}{t}, \quad d = st, \quad t = \frac{d}{s}$$

Density
Measurement · 5·p15

$$\rho = \frac{m}{V}, \quad m = \rho V, \quad V = \frac{m}{\rho}$$

counting a complement
Sets · 6·p3

$$n(A') = n(\xi) - n(A)$$

counting a union (don't count the overlap twice)
Sets · 6·p5

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

De Morgan's law for sets
Sets · 6·p9

$$(A \cup B)' = A' \cap B'$$

De Morgan's law for sets
Sets · 6·p9

$$(A \cap B)' = A' \cup B'$$

subsets of a set with n elements
Sets · 6·p10

$$\text{number of subsets} = 2^n$$

size of the power set
Sets · 6·p10

$$|P(S)| = 2^{|S|}$$

Three sets
Sets · 6·p13

$$n(A \cup B \cup C) = \text{singles} - \text{pairs} + n(A \cap B \cap C)$$

rows in a truth table with n letters
Logic · 7·p5

$$\text{rows} = 2^n$$

De Morgan's law
Logic · 7·p7

$$\neg(A \wedge B) = \neg A \vee \neg B$$

De Morgan's law
Logic · 7·p7

$$\neg(A \vee B) = \neg A \wedge \neg B$$

implication as OR
Logic · 7·p8

$$P \Rightarrow Q \equiv \neg P \vee Q$$

negating an implication
Logic · 7·p8

$$\neg(P \Rightarrow Q) \equiv P \wedge \neg Q$$

if and only if
Logic · 7·p10

$$P \Leftrightarrow Q \equiv (P \Rightarrow Q) \wedge (Q \Rightarrow P)$$

negating 'for all'
Logic · 7·p12

$$\neg\, \forall x\, P(x) \equiv \exists x\, \neg P(x)$$

negating 'there exists'
Logic · 7·p12

$$\neg\, \exists x\, P(x) \equiv \forall x\, \neg P(x)$$

Part 2: Algebra

Hidden ×
Algebra · 8·p13

$$3x = 3 \times x, \quad xy = x \times y$$

Division
Algebra · 8·p13

$$\frac{x}{2} = x \div 2$$

Powers
Algebra · 8·p13

$$3x^2 = 3 \times x \times x, \quad (3x)^2 = 9x^2$$

Formula / identity
Algebra · 8·p13

$$2(x+3) \equiv 2x + 6$$

Sign rules
Algebra · 8·p13

$$(-) \times (-) = +, \quad (+) \times (-) = -$$

Powers of negative numbers
Algebra · 8·p13

$$(-3)^2 = 9, \quad -3^2 = -9, \quad (-2)^3 = -8$$

multiplying powers of the same base
Indices (powers) · 9·p5

$$a^m \times a^n = a^{m+n}$$

dividing powers of the same base
Indices (powers) · 9·p6

$$\frac{a^m}{a^n} = a^{m-n}$$

power of a power
Indices (powers) · 9·p7

$$(a^m)^n = a^{mn}$$

power of a product
Indices (powers) · 9·p7
$$(ab)^n = a^n b^n$$

zero power
Indices (powers) · 9·p8
$$a^0 = 1$$

negative power
Indices (powers) · 9·p8
$$a^{-n} = \frac{1}{a^n}$$

fractional power
Indices (powers) · 9·p10
$$a^{\frac{m}{n}} = \left(\sqrt[n]{a}\right)^m$$

Power
Indices (powers) · 9·p18
$$a^n = a \times a \times \cdots \times a$$

Brackets
Indices (powers) · 9·p18
$$(ab)^n = a^n b^n, \quad \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

Zero power
Indices (powers) · 9·p18
$$a^0 = 1 \quad (a \neq 0)$$

Negative power
Indices (powers) · 9·p18
$$a^{-n} = \frac{1}{a^n}, \quad \frac{1}{a^{-n}} = a^n$$

Negative power of a fraction
Indices (powers) · 9·p18
$$\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n$$

Fraction power
Indices (powers) · 9·p18
$$a^{\frac{1}{n}} = \sqrt[n]{a}, \quad a^{\frac{m}{n}} = \left(\sqrt[n]{a}\right)^m$$

Negative base
Indices (powers) · 9·p18
$$(-2)^4 = 16, \ (-2)^3 = -8, \ -3^2 = -9$$

Roots of negatives
Indices (powers) · 9·p18
$$\sqrt[3]{-8} = -2$$

Unknown power
Indices (powers) · 9·p18
$$a^p = a^q \Rightarrow p = q$$

Unknown base
Indices (powers) · 9·p18
$$x^{\frac{m}{n}} = c \Rightarrow x = c^{\frac{n}{m}}$$

Standard form
Indices (powers) · 9·p18
$$a \times 10^n, \quad 1 \leq a < 10$$

Place value in base b
Number bases · 10·p14
$$(d_2\, d_1\, d_0)_b = d_2 b^2 + d_1 b + d_0$$

Binary addition
Number bases · 10·p14
$$1 + 1 = 10_2, \quad 1 + 1 + 1 = 11_2$$

Binary subtraction
Number bases · 10·p14
$$10_2 - 1 = 1$$

Bits and bytes
Number bases · 10·p14

$$n \text{ bits} \to 2^n \text{ values}$$

expanding a bracket
Simplifying · 11·p5

$$a(b+c) = ab + ac$$

expanding two brackets
Simplifying · 11·p6

$$(a+b)(c+d) = ac + ad + bc + bd$$

perfect square
Simplifying · 11·p8

$$(a+b)^2 = a^2 + 2ab + b^2$$

perfect square (minus)
Simplifying · 11·p8

$$(a-b)^2 = a^2 - 2ab + b^2$$

difference of two squares
Simplifying · 11·p8

$$(a-b)(a+b) = a^2 - b^2$$

Collecting
Simplifying · 11·p12

$$3x^2 - 5x^2 = -2x^2$$

Unlike terms
Simplifying · 11·p12

$$x + x^2, \quad 7a + 3b$$

Multiplying terms
Simplifying · 11·p12

$$-4x \times 3x^2 = -12x^3$$

Sign rules
Simplifying · 11·p12

$$(+)(+) = +, \ (-)(-) = +, \ (+)(-) = -$$

Minus outside
Simplifying · 11·p12

$$-(x-3) = -x + 3$$

Perfect squares
Simplifying · 11·p12

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Difference of two squares
Simplifying · 11·p12

$$(a+b)(a-b) = a^2 - b^2$$

Dividing by x
Simplifying · 11·p12

$$\frac{2x+3}{x} = 2 + \frac{3}{x}$$

factorising x² + bx + c
Factorisation · 12·p5

$$x^2 + bx + c = (x+p)(x+q), \quad p+q = b, \ pq = c$$

the pair for factorising ax² + bx + c
Factorisation · 12·p7

$$m, \ n = \frac{b \pm \sqrt{b^2 - 4ac}}{2}$$

the grid formula (m, n the pair)
Factorisation · 12·p7

$$ax^2 + bx + c = \left(x + \frac{n}{a}\right)(ax + m)$$

difference of two squares (factorising)
Factorisation · 12·p11

$$a^2 - b^2 = (a - b)(a + b)$$

perfect squares (factorising)
Factorisation · 12·p12

$$a^2 \pm 2ab + b^2 = (a \pm b)^2$$

difference of cubes
Factorisation · 12·p13

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

sum of cubes
Factorisation · 12·p13

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

x² + bx + c
Factorisation · 12·p15

$$(x + p)(x + q): \quad p + q = b, \quad pq = c$$

ax² + bx + c (split the middle)
Factorisation · 12·p15

$$pq = ac, \quad p + q = b: \quad ax^2 + px + qx + c$$

Grid formula (alternative)
Factorisation · 12·p15

$$m,\ n = \frac{b \pm \sqrt{b^2 - 4ac}}{2}, \qquad \left(x + \frac{n}{a}\right)(ax + m)$$

Does it factorise?
Factorisation · 12·p15

$$\Delta = b^2 - 4ac$$

Grouping (4 terms)
Factorisation · 12·p15

$$ax + ay + bx + by = (a + b)(x + y)$$

adding / subtracting fractions
Algebraic fractions · 13·p9

$$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$$

Simplify
Algebraic fractions · 13·p15

$$\frac{x^2 + 3x}{x + 3} = \frac{x\cancel{(x + 3)}}{\cancel{(x + 3)}} = x$$

Back-to-front brackets
Algebraic fractions · 13·p15

$$a - b = -(b - a), \qquad \frac{a - b}{b - a} = -1$$

Subtracting
Algebraic fractions · 13·p15

$$\frac{x + 3}{2} - \frac{x - 1}{3} = \frac{3(x + 3) - 2(x - 1)}{6}$$

Multiply
Algebraic fractions · 13·p15

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

Divide
Algebraic fractions · 13·p15

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

Splitting
Algebraic fractions · 13·p15

$$\frac{a+b}{c} = \frac{a}{c} + \frac{b}{c}, \qquad \frac{a}{b+c} \neq \frac{a}{b} + \frac{a}{c}$$

Whole number with a fraction
Algebraic fractions · 13·p15

$$1 - \frac{3}{x+1} = \frac{x+1-3}{x+1}$$

Undo buttons
Transposing formulae · 14·p21

$$+ \leftrightarrow -, \quad \times \leftrightarrow \div, \quad x^2 \leftrightarrow \pm\sqrt{x}, \quad x^3 \leftrightarrow \sqrt[3]{x}$$

Minus in front
Transposing formulae · 14·p21

$$y = 5 - 2x \;\Rightarrow\; 2x = 5 - y$$

Letter on the bottom
Transposing formulae · 14·p21

$$P = \frac{V}{R} \;\Rightarrow\; PR = V \;\Rightarrow\; R = \frac{V}{P}$$

Flipping
Transposing formulae · 14·p21

$$\frac{a}{b} = \frac{c}{d} \;\Rightarrow\; \frac{b}{a} = \frac{d}{c}$$

Letter twice
Transposing formulae · 14·p21

$$yx - x = 1 + y \;\Rightarrow\; x(y-1) = 1 + y$$

Roots
Transposing formulae · 14·p21

$$r^2 = k \Rightarrow r = \pm\sqrt{k}, \qquad r^3 = k \Rightarrow r = \sqrt[3]{k}$$

quadratic formula (for $ax^2 + bx + c = 0$)
Solving equations · 15·p12

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

completing the square
Solving equations · 15·p13

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 + c$$

Cross-multiplying
Solving equations · 15·p18

$$\frac{a}{b} = \frac{c}{d} \;\Rightarrow\; ad = bc$$

Zero product
Solving equations · 15·p18

$$AB = 0 \;\Rightarrow\; A = 0 \text{ or } B = 0$$

Discriminant
Solving equations · 15·p18

$$\Delta = b^2 - 4ac$$

inverse of a 2 × 2 matrix
Matrices · 16·p8

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Size
Matrices · 16·p16

$$m \times n$$

Scalar multiple
Matrices · 16·p16

$$k \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$$

Size rule
Matrices · 16·p16

$$(m \times n)(n \times p) = m \times p$$

Order
Matrices · 16·p16

$$AB \neq BA$$

Identity
Matrices · 16·p16

$$AI = IA = A$$

Determinant
Matrices · 16·p16

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

Solving AX = B
Matrices · 16·p16

$$X = A^{-1}B$$

Undoing two moves
Matrices · 16·p16

$$(TS)^{-1} = S^{-1}T^{-1}$$

nth term of an arithmetic progression
Sequences and series · 17·p3

$$u_n = a + (n-1)d$$

sum of an arithmetic progression (first a, last l, n terms)
Sequences and series · 17·p4

$$S_n = \frac{(a+l)}{2}n$$

sum of an arithmetic progression (n terms, difference d)
Sequences and series · 17·p5

$$S_n = \frac{n}{2}\big(2a + (n-1)d\big)$$

nth term of a geometric progression
Sequences and series · 17·p7

$$u_n = ar^{n-1}$$

sum of the first n terms of a geometric progression
Sequences and series · 17·p8

$$S_n = \frac{a(1-r^n)}{1-r}$$

sum to infinity of a geometric progression
Sequences and series · 17·p10

$$S_\infty = \frac{a}{1-r} \quad (|r| < 1)$$

Number of terms
Sequences and series · 17·p14

$$n = \frac{l-a}{d} + 1$$

Sigma
Sequences and series · 17·p14

$$\sum_{i=1}^{n} f(i) = f(1) + f(2) + \cdots + f(n)$$

Odd numbers
Sequences and series · 17·p14

$$1 + 3 + 5 + \cdots + (2n-1) = n^2$$

Part 3: Functions and graphs

Notation
Functions · 18·p13
$f(a) = \text{the rule with every } x \text{ replaced by } (a)$

Composite
Functions · 18·p13
$fg(x) = (f \circ g)(x) = f(g(x))$

Inverse
Functions · 18·p13
$f(f^{-1}(x)) = f^{-1}(f(x)) = x$

Coordinates
Graphs of functions · 19·p13
$(x,\ y)$

Even function
Graphs of functions · 19·p13
$f(-x) = f(x)$

Odd function
Graphs of functions · 19·p13
$f(-x) = -f(x)$

Asymptotes
Graphs of functions · 19·p13
$y = \dfrac{b}{x-h} + k: \quad x = h,\ y = k$

Vertex form
Graphs of functions · 19·p13
$y = a(x-h)^2 + k \ \Rightarrow\ \text{vertex } (h, k)$

Shift up / down
Graphs of functions · 19·p13
$y = f(x) + a$

Shift left / right
Graphs of functions · 19·p13
$y = f(x + a)$

Vertical stretch
Graphs of functions · 19·p13
$y = af(x)$

Horizontal squash
Graphs of functions · 19·p13
$y = f(ax)$

Reflections
Graphs of functions · 19·p13
$y = -f(x), \quad y = f(-x)$

gradient through two points
The straight line · 20·p4
$m = \dfrac{y_2 - y_1}{x_2 - x_1}$

line with gradient m through (x_1, y_1)
The straight line · 20·p5
$y - y_1 = m(x - x_1)$

parallel lines
The straight line · 20·p9
$m_1 = m_2$

perpendicular lines
The straight line · 20·p9
$m_1 m_2 = -1 \quad \Leftrightarrow \quad m_2 = -\dfrac{1}{m_1}$

midpoint
The straight line · 20·p16

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

distance between two points
The straight line · 20·p16

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Straight line
The straight line · 20·p21

$$y = mx + c$$

Horizontal / vertical
The straight line · 20·p21

$$y = b \ \ (m = 0), \qquad x = a \ \ (m \text{ undefined})$$

Perpendicular
The straight line · 20·p21

$$m_1 m_2 = -1, \qquad m_2 = -\frac{1}{m_1}$$

Midpoint
The straight line · 20·p21

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Distance
The straight line · 20·p21

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Collinear
The straight line · 20·p21

$$m_{AB} = m_{BC}$$

Exponential model
The exponential function · 21·p11

$$y = A \times b^x$$

e form
The exponential function · 21·p11

$$y = Ae^{kx}$$

Percentage change
The exponential function · 21·p11

$$\times \left(1 \pm \frac{r}{100}\right) \text{ per step}$$

Compound interest
The exponential function · 21·p11

$$A\left(1 + \frac{r}{n}\right)^{nt}$$

The number e
The exponential function · 21·p11

$$e = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n \approx 2.718$$

Levelling off
The exponential function · 21·p11

$$y = A + Be^{-kt} \ \rightarrow \ A$$

Time constant
The exponential function · 21·p11

$$i = Ie^{-t/\tau}$$

Half-life
The exponential function · 21·p11

$$y = A\left(\tfrac{1}{2}\right)^{t/T}$$

Doubling time
The exponential function · 21·p11

$$y = A \times 2^{t/T}$$

log law: product
Logarithms · 22·p5

$$\log(ab) = \log a + \log b$$

log law: quotient
Logarithms · 22·p5

$$\log \frac{a}{b} = \log a - \log b$$

log law: power
Logarithms · 22·p5

$$\log a^n = n \log a$$

change of base
Logarithms · 22·p7

$$\log_a N = \frac{\log_b N}{\log_b a}$$

Definition
Logarithms · 22·p13

$$y = \log_b x \iff b^y = x$$

Special values
Logarithms · 22·p13

$$\log_b b = 1, \qquad \log_b 1 = 0$$

Undo
Logarithms · 22·p13

$$\log_b(b^x) = x, \qquad b^{\log_b x} = x$$

ln and e
Logarithms · 22·p13

$$\ln(e^x) = x, \quad e^{\ln x} = x, \quad \ln e = 1, \quad \ln 1 = 0$$

Reciprocal
Logarithms · 22·p13

$$\log \frac{1}{a} = -\log a$$

Part 4: Data and chance

Pie chart angle
Tables and charts · 23·p14

$$\text{angle} = \frac{\text{frequency}}{\text{total}} \times 360°$$

Pie chart frequency
Tables and charts · 23·p14

$$\text{frequency} = \frac{\text{angle}}{360°} \times \text{total}$$

Frequency density
Tables and charts · 23·p14

$$\text{frequency density} = \frac{\text{frequency}}{\text{class width}}$$

Frequency from a histogram
Tables and charts · 23·p14

$$\text{frequency} = \text{frequency density} \times \text{class width}$$

mean
Statistics · 24·p2

$$\bar{x} = \frac{\sum x_i}{n}$$

population variance
Statistics · 24·p11

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

sample standard deviation
Statistics · 24·p12

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

Mean
Statistics · 24·p15

$$\bar{x} = \frac{\sum x}{n}$$

Total from a mean
Statistics · 24·p15

$$\text{total} = n \times \bar{x}$$

Mean of a frequency table
Statistics · 24·p15

$$\bar{x} = \frac{\sum fx}{\sum f}$$

Median position
Statistics · 24·p15

$$\frac{n + 1}{2}$$

Quartile position
Statistics · 24·p15

$$\frac{\text{median position} + 1}{2}$$

Fractional position
Statistics · 24·p15

$$k + d \;\rightarrow\; x_k + d\,(x_{k+1} - x_k)$$

Range
Statistics · 24·p15

$$\text{largest} - \text{smallest}$$

Interquartile range
Statistics · 24·p15

$$\text{IQR} = \text{UQ} - \text{LQ}$$

Population variance and SD
Statistics · 24·p15

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N}, \qquad \sigma = \sqrt{\sigma^2}$$

Sample SD
Statistics · 24·p15

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

addition rule (OR)
Probability and Bayes · 25·p3

$$p(E \cup F) = p(E) + p(F) - p(E \cap F)$$

conditional probability
Probability and Bayes · 25·p4

$$p(E \mid F) = \frac{p(E \cap F)}{p(F)}$$

Bayes' theorem
Probability and Bayes · 25·p10

$$p(a \mid b) = \frac{p(b \mid a)\, p(a)}{p(b)}$$

law of total probability
Probability and Bayes · 25·p11

$$p(B) = p(B \mid A)\, p(A) + p(B \mid A')\, p(A')$$

Probability
Probability and Bayes · 25·p15

$$p(E) = \frac{\text{outcomes in } E}{\text{all outcomes}}$$

Complement
Probability and Bayes · 25·p15

$$p(E') = 1 - p(E)$$

Addition rule (OR)
Probability and Bayes · 25·p15

$$p(A \cup B) = p(A) + p(B) - p(A \cap B)$$

Mutually exclusive
Probability and Bayes · 25·p15

$$p(A \cap B) = 0, \qquad p(A \cup B) = p(A) + p(B)$$

Conditional
Probability and Bayes · 25·p15

$$p(A \mid B) = \frac{p(A \cap B)}{p(B)}$$

Multiplication rule (AND)
Probability and Bayes · 25·p15

$$p(A \cap B) = p(B)\, p(A \mid B) = p(A)\, p(B \mid A)$$

Independent
Probability and Bayes · 25·p15

$$p(A \cap B) = p(A)\, p(B), \qquad p(A \mid B) = p(A)$$

Bayes' theorem
Probability and Bayes · 25·p15

$$p(A \mid B) = \frac{p(B \mid A)\, p(A)}{p(B)}$$

Normal curve
The Normal distribution · 26·p4

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

z-score
The Normal distribution · 26·p6

$$z = \frac{x - \mu}{\sigma}$$

Back from z
The Normal distribution · 26·p15

$$x = \mu + z\sigma, \qquad \mu = x - z\sigma, \qquad \sigma = \frac{x - \mu}{z}$$

Below a value
The Normal distribution · 26·p15

$$p(X < x) = \Phi(z)$$

Above a value
The Normal distribution · 26·p15

$$p(X > x) = 1 - \Phi(z)$$

Between two values
The Normal distribution · 26·p15

$$p(a < X < b) = \Phi(z_b) - \Phi(z_a)$$

Negative z
The Normal distribution · 26·p15

$$\Phi(-z) = 1 - \Phi(z), \qquad \Phi(0) = 0.5$$

Part 5: Geometry and trigonometry

degrees and radians
Angles · 27·p5
$$360^\circ = 2\pi \text{ rad}$$

degrees to radians
Angles · 27·p6
$$\text{rad} = \text{deg} \times \frac{\pi}{180}$$

arc length (θ in degrees)
Angles · 27·p8
$$s = \frac{\theta}{360} \times 2\pi r$$

sector area (θ in degrees)
Angles · 27·p8
$$A = \frac{\theta}{360} \times \pi r^2$$

arc length (θ in radians)
Angles · 27·p8
$$s = r\theta$$

sector area (θ in radians)
Angles · 27·p8
$$A = \tfrac{1}{2} r^2 \theta$$

Degrees → radians
Angles · 27·p10
$$\theta_{\text{rad}} = \theta_{\text{deg}} \times \frac{\pi}{180}$$

Radians → degrees
Angles · 27·p10
$$\theta_{\text{deg}} = \theta_{\text{rad}} \times \frac{180}{\pi}$$

Arc length
Angles · 27·p10
$$s = r\theta \quad \text{or} \quad s = \frac{\theta}{360} \times 2\pi r$$

Sector area
Angles · 27·p10
$$A = \tfrac{1}{2} r^2 \theta \quad \text{or} \quad A = \frac{\theta}{360} \times \pi r^2$$

Pythagoras
Intro to trigonometry · 28·p3
$$h^2 = o^2 + a^2$$

circle, centre (a, b), radius r
Intro to trigonometry · 28·p5
$$(x - a)^2 + (y - b)^2 = r^2$$

sine
Intro to trigonometry · 28·p6
$$\sin\theta = \frac{o}{h}$$

cosine
Intro to trigonometry · 28·p6
$$\cos\theta = \frac{a}{h}$$

tangent
Intro to trigonometry · 28·p6
$$\tan\theta = \frac{o}{a}$$

Pythagoras
Intro to trigonometry · 28·p13

$$a^2 + b^2 = c^2$$

Pythagoras in 3D
Intro to trigonometry · 28·p13

$$d^2 = a^2 + b^2 + c^2$$

SOHCAHTOA
Intro to trigonometry · 28·p13

$$\sin\theta = \frac{O}{H}, \quad \cos\theta = \frac{A}{H}, \quad \tan\theta = \frac{O}{A}$$

Inverse trig
Intro to trigonometry · 28·p13

$$\theta = \sin^{-1}\frac{O}{H},\ \cos^{-1}\frac{A}{H},\ \tan^{-1}\frac{O}{A}$$

Exact values
Intro to trigonometry · 28·p13

$$\sin 30^\circ = \tfrac{1}{2},\ \sin 45^\circ = \tfrac{1}{\sqrt{2}},\ \sin 60^\circ = \tfrac{\sqrt{3}}{2}$$

general sine wave
Trig graphs · 29·p8

$$y = A\sin(B(x-C)) + D:\ \text{amplitude } |A|,\ \text{period } \frac{2\pi}{B},\ \text{shift } C,\ \text{midline } D$$

stretching trig graphs
Trig graphs · 29·p8

$$y = A\sin(Bx):\ \text{amplitude } A,\ \text{period } \frac{2\pi}{B}$$

sine series
Trig graphs · 29·p13

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

cosine series
Trig graphs · 29·p13

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

tangent series
Trig graphs · 29·p15

$$\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots \quad (|x| < \tfrac{\pi}{2})$$

Unit circle
Trig graphs · 29·p17

$$(\cos\theta,\ \sin\theta), \qquad \tan\theta = \frac{\sin\theta}{\cos\theta}$$

Pythagorean identity
Trig graphs · 29·p17

$$\cos^2\theta + \sin^2\theta = 1$$

General wave
Trig graphs · 29·p17

$$y = A\sin(B(x-C)) + D$$

tan(Bx)
Trig graphs · 29·p17

$$\text{period} = \frac{180^\circ}{B} = \frac{\pi}{B}$$

Read off a graph
Trig graphs · 29·p17

$$D = \frac{\max + \min}{2}, \quad A = \frac{\max - \min}{2}, \quad B = \frac{360^\circ}{\text{period}}$$

Power series
Trig graphs · 29·p17

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots, \quad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

Pythagorean identity
Trig identities and equations · 30·p2

$$\sin^2\theta + \cos^2\theta = 1$$

r sin form
Trig identities and equations · 30·p8

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha),\ r = \sqrt{a^2+b^2},\ \alpha = \tan^{-1}\frac{b}{a}$$

tan
Trig identities and equations · 30·p11

$$\tan\theta = \frac{\sin\theta}{\cos\theta}, \qquad \tan^2\theta + 1 = \frac{1}{\cos^2\theta}$$

Partner: sin
Trig identities and equations · 30·p11

$$\theta_1, \quad 180^\circ - \theta_1 \quad (\pi - \theta_1)$$

Partner: cos
Trig identities and equations · 30·p11

$$\theta_1, \quad 360^\circ - \theta_1 \quad (2\pi - \theta_1)$$

Partner: tan
Trig identities and equations · 30·p11

$$\theta_1, \quad \theta_1 + 180^\circ \quad (\theta_1 + \pi)$$

r sin form
Trig identities and equations · 30·p11

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Compound angle
Trig identities and equations · 30·p11

$$\sin(A+B) = \sin A\cos B + \cos A\sin B$$

sine rule
Solving triangles · 31·p3

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

cosine rule (angle form)
Solving triangles · 31·p5

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

cosine rule
Solving triangles · 31·p5

$$a^2 = b^2 + c^2 - 2bc\cos A$$

area of a triangle
Solving triangles · 31·p9

$$\text{Area} = \tfrac{1}{2}bc\sin A$$

Angle sum
Solving triangles · 31·p12

$$A + B + C = 180^\circ$$

Area
Solving triangles · 31·p12

$$\text{Area} = \tfrac{1}{2}ab\sin C$$

Area backwards
Solving triangles · 31·p12

$$\sin C = \frac{2\times\text{Area}}{ab}, \qquad a = \frac{2\times\text{Area}}{b\sin C}$$

magnitude (length) of a vector
Vectors · 32·p6

$$|\mathbf{a}| = \sqrt{x^2 + y^2 + z^2}$$

unit vector in the direction of a
Vectors · 32·p7

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}$$

position vectors: A to B, and the midpoint
Vectors · 32·p7

$$\overrightarrow{AB} = \mathbf{b} - \mathbf{a}, \qquad \mathbf{m} = \frac{\mathbf{a} + \mathbf{b}}{2}$$

dot product (angle form)
Vectors · 32·p9

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta$$

angle between two vectors
Vectors · 32·p10

$$\cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$$

dot product (components)
Vectors · 32·p10

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z$$

size of the cross product
Vectors · 32·p13

$$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin\theta$$

cross product (components)
Vectors · 32·p16

$$\mathbf{a} \times \mathbf{b} = \mathbf{i}(a_y b_z - a_z b_y) - \mathbf{j}(a_x b_z - a_z b_x) + \mathbf{k}(a_x b_y - a_y b_x)$$

Components
Vectors · 32·p17

$$\mathbf{a} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

Direction angle
Vectors · 32·p17

$$\theta = \tan^{-1}\frac{y}{x}$$

Dot product
Vectors · 32·p17

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z = |\mathbf{a}||\mathbf{b}| \cos\theta$$

polar to Cartesian
Polar coordinates · 33·p3

$$x = r\cos\theta, \quad y = r\sin\theta$$

Cartesian to polar
Polar coordinates · 33·p4

$$r = \sqrt{x^2 + y^2}, \quad \theta = \tan^{-1}\frac{y}{x} \ (+\text{quadrant fix})$$

polar and Cartesian links
Polar coordinates · 33·p7

$$x^2 + y^2 = r^2, \qquad \tan\theta = \frac{y}{x}$$

double angle
Polar coordinates · 33·p10

$$\sin 2A = 2\sin A\cos A, \quad \cos 2A = \cos^2 A - \sin^2 A$$

Polar → Cartesian
Polar coordinates · 33·p11

$$x = r\cos\theta, \qquad y = r\sin\theta$$

Distance
Polar coordinates · 33·p11

$$r = \sqrt{x^2 + y^2}$$

Angle
Polar coordinates · 33·p11

$$\theta = \tan^{-1}\frac{y}{x}$$

Double angle
Polar coordinates · 33·p11

$$\sin 2A = 2\sin A\cos A, \quad \cos 2A = \cos^2 A - \sin^2 A$$

Part 6: Calculus

power rule
Gradients of curves · 34·p5

$$\frac{d}{dx}\left(ax^n\right) = anx^{n-1}$$

Notation
Gradients of curves · 34·p18

$$\frac{dy}{dx} = f'(x) = \dot{f}(x)$$

Constants, kx, sums
Gradients of curves · 34·p18

$$\frac{d}{dx}(c) = 0, \qquad \frac{d}{dx}(kx) = k$$

Rewrite first
Gradients of curves · 34·p18

$$\frac{a}{x^n} = ax^{-n}, \quad \sqrt{x} = x^{\frac{1}{2}}, \quad \frac{1}{\sqrt{x}} = x^{-\frac{1}{2}}$$

ln and e
Gradients of curves · 34·p18

$$\frac{d}{dx}\ln x = \frac{1}{x}, \qquad \frac{d}{dx}e^{kx} = ke^{kx}$$

Trig (radians)
Gradients of curves · 34·p18

$$\frac{d}{dx}\sin bx = b\cos bx, \quad \frac{d}{dx}\cos bx = -b\sin bx, \quad \frac{d}{dx}\tan x = \sec^2 x$$

Gradient at x = a
Gradients of curves · 34·p18

$$m = f'(a)$$

Tangent at x = a
Gradients of curves · 34·p18

$$y - y_1 = f'(a)(x - x_1)$$

Normal at x = a
Gradients of curves · 34·p18

$$m_N = -\frac{1}{f'(a)}$$

Stationary points
Gradients of curves · 34·p18

$$\frac{dy}{dx} = 0$$

Second-derivative test
Gradients of curves · 34·p18

$$\frac{d^2y}{dx^2} > 0 \text{ min}, \quad < 0 \text{ max}, \quad = 0 \text{ test fails}$$

chain rule
Chain, product, quotient · 35·p4

$$\frac{dy}{dx} = \frac{dy}{dz} \times \frac{dz}{dx}$$

product rule
Chain, product, quotient · 35·p8

$$\frac{d(uv)}{dx} = v\frac{du}{dx} + u\frac{dv}{dx}$$

quotient rule
Chain, product, quotient · 35·p10

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

Higher derivatives
Chain, product, quotient · 35·p17

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$$

Chain rule
Chain, product, quotient · 35·p17

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

Chain: powers
Chain, product, quotient · 35·p17

$$\frac{d}{dx}\big(g(x)\big)^n = n\big(g(x)\big)^{n-1} g'(x)$$

Chain: e and ln
Chain, product, quotient · 35·p17

$$\frac{d}{dx}e^{g(x)} = g'(x)e^{g(x)}, \qquad \frac{d}{dx}\ln g(x) = \frac{g'(x)}{g(x)}$$

Chain: roots
Chain, product, quotient · 35·p17

$$\frac{d}{dx}\sqrt{g(x)} = \frac{g'(x)}{2\sqrt{g(x)}}$$

Constant multiple
Chain, product, quotient · 35·p17

$$\frac{d}{dx}\big(kf(x)\big) = kf'(x)$$

trapezium rule
Integration and areas · 36·p3

$$\int_a^b y\,dx \approx \frac{h}{2}\big(y_0 + 2(y_1 + \cdots + y_{n-1}) + y_n\big), \; h = \frac{b-a}{n}$$

power rule for integration
Integration and areas · 36·p6

$$\int ax^n\,dx = \frac{a}{n+1}x^{n+1} + c \quad (n \neq -1)$$

integral of 1/x
Integration and areas · 36·p9

$$\int \frac{1}{x}\,dx = \ln|x| + c$$

definite integral
Integration and areas · 36·p13

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

Constant
Integration and areas · 36·p18

$$\int k\,dx = kx + c$$

Rewrite first
Integration and areas · 36·p18

$$\frac{3}{x^2} = 3x^{-2}, \qquad \sqrt{x} = x^{\frac{1}{2}}, \qquad \frac{1}{\sqrt{x}} = x^{-\frac{1}{2}}$$

Sums and multiples
Integration and areas · 36·p18

$$\int (f \pm g)\,dx = \int f\,dx \pm \int g\,dx, \qquad \int kf\,dx = k\int f\,dx$$

e and trig
Integration and areas · 36·p18

$$\int e^{kx}dx = \tfrac{1}{k}e^{kx}, \quad \int \sin kx\,dx = -\tfrac{1}{k}\cos kx, \quad \int \cos kx\,dx = \tfrac{1}{k}\sin kx$$

tan
Integration and areas · 36·p18

$$\int \tan x \, dx = \ln|\sec x| + c$$

Trapezium rule
Integration and areas · 36·p18

$$\frac{h}{2}\big(y_0 + 2(y_1 + \cdots + y_{n-1}) + y_n\big), \quad h = \frac{b-a}{n}$$

integration by parts
Parts and substitution · 37·p5

$$\int u\frac{dv}{dx}\,dx = uv - \int v\frac{du}{dx}\,dx$$

definite integration by parts
Parts and substitution · 37·p10

$$\int_a^b u\frac{dv}{dx}\,dx = \Big[uv\Big]_a^b - \int_a^b v\frac{du}{dx}\,dx$$

integration by substitution (y = g(x))
Parts and substitution · 37·p16

$$\int_a^b f(g(x))\,g'(x)\,dx = \int_{g(a)}^{g(b)} f(y)\,dy$$

linear inside shortcut
Parts and substitution · 37·p17

$$\int f(ax+b)\,dx = \frac{1}{a}F(ax+b) + C$$

ln x
Parts and substitution · 37·p20

$$\int \ln x \, dx = x\ln x - x + C$$

small changes (two variables)
Partial differentiation · 38·p11

$$\delta z \approx \frac{\partial z}{\partial x}\delta x + \frac{\partial z}{\partial y}\delta y$$

Partial derivative
Partial differentiation · 38·p13

$$\frac{\partial f}{\partial x} = f_x$$

Second partials
Partial differentiation · 38·p13

$$f_{xx} = \frac{\partial^2 f}{\partial x^2}, \qquad f_{xy} = \frac{\partial^2 f}{\partial y\,\partial x}$$

Mixed partials
Partial differentiation · 38·p13

$$\frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial^2 f}{\partial x\,\partial y}$$

Gradient vector
Partial differentiation · 38·p13

$$\nabla f = \begin{pmatrix} f_x \\ f_y \end{pmatrix}$$

Standard derivatives and integrals

Chapters 34–37

Read across: differentiate a function to get the middle column; integrate it to get the right-hand column. Angles are in **radians** whenever you differentiate or integrate trig functions.

Function $f(x)$	Derivative $f'(x)$	Integral $\int f(x)\,dx$
k (a constant)	0	$kx + C$
x^n	nx^{n-1}	$\dfrac{x^{n+1}}{n+1} + C \quad (n \neq -1)$
$\dfrac{1}{x} = x^{-1}$	$-\dfrac{1}{x^2}$	$\ln\lvert x\rvert + C$
e^x	e^x	$e^x + C$
e^{kx}	$k\,e^{kx}$	$\dfrac{1}{k}e^{kx} + C$
a^x	$a^x \ln a$	$\dfrac{a^x}{\ln a} + C$
$\ln x$	$\dfrac{1}{x}$	$x\ln x - x + C$
$\sin x$	$\cos x$	$-\cos x + C$
$\cos x$	$-\sin x$	$\sin x + C$
$\tan x$	$\sec^2 x$	$\ln\lvert\sec x\rvert + C$
$\sin kx$	$k\cos kx$	$-\dfrac{1}{k}\cos kx + C$
$\cos kx$	$-k\sin kx$	$\dfrac{1}{k}\sin kx + C$

Calculus rules

Chapters 34–37

Sum rule: do each term separately

$$(f \pm g)' = f' \pm g', \qquad \int (f \pm g) = \int f \pm \int g$$

A constant multiple stays put (not a product rule case)

$$(kf)' = kf', \qquad \int kf = k\int f$$

Product rule

$$\frac{\mathrm{d}}{\mathrm{d}x}(uv) = u\frac{\mathrm{d}v}{\mathrm{d}x} + v\frac{\mathrm{d}u}{\mathrm{d}x}$$

Quotient rule

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{u}{v}\right) = \frac{v\,u' - u\,v'}{v^2}$$

Chain rule (function of a function)

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}u} \cdot \frac{\mathrm{d}u}{\mathrm{d}x}$$

Linear inside: integrate, then divide by a

$$\int f(ax+b)\,\mathrm{d}x = \frac{1}{a}F(ax+b) + C$$

Integration by parts

$$\int u\,\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = uv - \int v\,\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$$

Definite integral

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a)$$

Answer Key and Tracker

The answers to every "Your turn" page, without the questions, so you can check a whole chapter at a glance. Each small circle is one question: tick it in Xournal++ when you get it right, and watch the book fill up.

1 Whole numbers

○○○ Bundles

1. 6 hundreds (600) 2. 125 (ones: 8+7 = 15, carry 1) 3. 45 (open a bundle: 11 - 6, 7 - 3)

○○○○ The lift

4. 5 5. -5 6. 3 (taking away a debt) 7. 20 (same signs: positive)

○○○ Getting dressed

8. 4 (multiply first) 9. 24 (brackets first) 10. 11 (power first: 9, then + 2)

○○○○ Dots and lights

11. 1, 2, 3, 6, 9, 18 12. No: 21 = 3 x 7 13. 24 14. 12

○○○○○○ Signs, order and primes

15. -15 (both negative: add sizes) 16. -8 (left to right: -4 × 2) 17. 25 and -25 18. 2 (inner bracket 3, then 10 - 9 = 1) 19. 3 (15 ÷ 5) 20. $18 = 2 \times 3^2$, $24 = 2^3 \times 3$: HCF = $2 \times 3 = 6$, LCM = $2^3 \times 3^2 = 72$

○○○○○○ Find the unknown

21. 4 (? = -3 + 7) 22. -7 (? = 28 ÷ (-4)) 23. -30 (? = 5 × (-6)) 24. -6 (? = 9 - 15) 25. 20 or 28 (multiples of 4; 24 is out because HCF(12, 24) = 12) 26. 5 (30 = 2 × 3 × 5 and 6 = 2 × 3, so n must supply the 5; 10 is too big)

2 Fractions

○○○○ Simplify and convert

1. 3/4 2. 4 3/5 3. 23/7 4. 7/11 (5×11 = 55 < 56 = 7×8)

○○○○○ Add and subtract

5. 7/9 6. 7/20 7. 29/24 = 1 5/24 8. −5/12 9. 23/15 = 1 8/15

○○○○○ Multiply and divide

10. 1/6 11. 7/2 = 3 1/2 12. 3/4 13. 1/8 14. 1/2

○○○○ Mixed operations

15. 1 16. 7/8 17. 27 18. −1/4

○○○○○ Find the unknown

19. 25 20. 5/8 21. 2/3 22. 24 23. 1/2

3 Decimals

○○○○ Reading and comparing

1. 6 hundredths 2. 0.7 3. 0.3, 0.303, 0.33 4. 7/20

○○○○○ Working with them

5. 16.15 6. 6.38 7. 0.42 8. 4.2 9. 90

○○○ Fraction to decimal

10. 0.625 11. 0.6 recurring 12. 0.35

○○○○○ Rounding

13. 6.7 14. 6.75 15. 0.030 16. 73,000 17. 10 × 50 = 500

○○○○○○ Significant figures, recurring, bounds

18. 3, 2, 3 19. 0.0410 (4, 0, 9; decider 6 bumps the 9) 20. 4/9 (10x − x = 4) 21. yes ($40 = 2^3 \times 5$); no (6 has a 3) 22. $12.55 \le x < 12.65$ 23. 8.5 ÷ 1.5 = 5.67 (upper ÷ lower)

○○○○○ Find the unknown

24. 0.9 (0.36 ÷ 0.4 = 3.6 ÷ 4) 25. 4.45 (2.65 + 1.8) 26. 3 (15 × 0.2) 27. $4.25 \le x < 4.35$ (± 0.05) 28. $0.555 \le x < 0.565$ (± 0.005)

4 Percentage and ratio

○○○○○ Percentages

1. 0.35 = 7/20 2. 7% 3. 13 4. 14 5. 40

○○○○ Up, down and change

6. 280 7. 68 8. -25% 9. 1.2 × 0.8 = 0.96 (a 4% fall)

○○○○ Ratio

10. 3 : 4 11. £36 and £48 12. 100 ml 13. 1 km

○ Mixtures

14. (60 + 15)/500 = 15%

○○○○○ Ratio rules

15. 1 : 4 (50 : 200) 16. 60%, 20%, 20% (9/15, 3/15, 3/15) 17. 2 : 3 (40 : 60) 18. 15 days (inverse: 6 × 10 = 60, ÷ 4) 19. A £49, B £14 (5 parts = £35, 1 part = £7)

○○○○○○ Find the unknown

20. £50 (55 ÷ 1.1) 21. £60 (45 ÷ 0.75) 22. 36 (5 parts = 20, 1 part = 4, 9 parts) 23. 10 (6x = 15 × 4 = 60) 24. 4 (5x = 2.5 × 8 = 20) 25. 150 (18 ÷ 0.12)

5 Measurement

○○○○○ Changing units

1. 3200 m 2. 0.45 m 3. 2500 g 4. 0.75 l 5. 4 ms

○○○ Shapes

6. 30 cm, 54 cm^2 7. 35 cm^2 8. 31.4 cm, 78.5 cm^2

○○○○ Area and volume units

9. 30,000 cm^2 10. 60 cm^3 11. 2000 cm^3 12. 500,000 cm^3

○○○ Time and per

13. 2 h 24 min 14. 60 km/h 15. 72 km/h

○○○○○○ More shapes, units and time

16. 28 cm² (½ × 14 × 4) 17. 88 m² (108 − 20) 18. 72 cm² (2(18 + 12 + 6)) 19. 6.5 cm³ (÷ 1000) 20. 3.67 h (3 + 40/60) 21. 10:25 (9 h 85 min = 10 h 25 min)

○○○○○○ Find the unknown

22. 8 cm (√64) 23. 9 m (54 ÷ 6) 24. 10.0 cm (62.8 ÷ 2π) 25. 7.0 cm (√(154 ÷ π)) 26. 2.5 h = 2 h 30 min (210 ÷ 84) 27. 5 cm (120 ÷ (6 × 4))

6 Sets

○○○○○○ Reading sets

1. no 2. {2, 4} 3. {1, 2, 3, 4, 5, 6, 8, 10} 4. {1, 3, 5} 5. {1, 3, 5, 7, 9} 6. 8

○○ Venn counting

7. 40 - (13 + 12 + 8) = 7 8. 13

○○○○ Decoding

9. {-2, -1, 0, 1, 2} 10. ℤ 11. yes (3/4); no 12. filled dot at 2, hollow at 5

○○○○○○ Set rules

13. {7, 9} 14. {6, 8, 10} 15. both {1, 3, 5, 6, 7, 8, 9, 10}; yes (De Morgan) 16. 2^4 = 16 17. yes; no (a proper subset must be smaller) 18. [4, ∞)

○○○○○○ Find the unknown

19. (28 − x) + x + (22 − x) + 9 = 50, so 59 − x = 50, x = 9 20. 25 + 21 − 40 = 6 21. 35 = 20 + n(B) − 8, n(B) = 23 22. 60 − 42 = 18 23. 39 − x = 32, x = 7 24. 2^n = 64, n = 6

○○ Three sets

25. 45 + 40 + 35 − 15 − 12 − 10 + 5 = 88 like at least one, so 100 − 88 = 12 26. 45 − (15 − 5) − (12 − 5) − 5 = 23

7 Logic

○○○ Statements

1. only once x has a value (true if x = 3) 2. T 3. F

○○○ Tables

4. T only in the row A = F, B = T 5. always T (called a tautology) 6. the columns match

○○ If... then

7. no (row 4) 8. converse: if even then divisible by 4 (false: 6); contrapositive: if odd then not divisible by 4 (true)

○○ Python

9. True 10. False

○○○○○○ Rules and bigger tables

11. 2^4 = 16 12. T F T F T F T F 13. (¬P) ∨ (Q ∧ R) 14. ¬P ∨ Q 15. some bus is not late 16. drive ⇒ licence (a licence is necessary, not sufficient)

○○○○○○ Find the unknown

17. P = T, Q = F (the only F row) 18. both F (the only F row of OR) 19. F (P = T would make T ⇒ F, which is F) 20. F (⇔ needs them to match) 21. F (P = T, Q = T, so T ⇒ F) 22. 2^n = 32, n = 5

8 Algebra

○○○ Reading algebra

1. 4 × a × b 2. no: 9 vs 6 3. 7, −1, 2

○○○○○○○○ Substitution (x = 3, y = −2)

4. 11 5. 11 6. 12 7. −10 8. 1 9. −8 10. 1/5 11. 5/(−4) = −5/4

○○○○ Words to algebra

12. 4n − 7 13. p/3 14. 5q + 90 15. 2n + 7

○ Types

16. (a) expression (b) equation (c) formula (d) identity

○○○○○○ Find the unknown

17. a = 5 (20 = 5 + 3a) 18. p = 5 (3p = 15) 19. w = 6 (30 = 18 + 2w) 20. b = 8 (24 = 3b) 21. 7 (4n − 7 = 21, 4n = 28) 22. k = 9 (6 + k = 15)

9 Indices (powers)

○○○○○○○○ Values

1. 64 2. 125 3. 1 4. 1/4 5. 1/9 6. 5 7. 9 8. 7

○○○○○○ Simplify

9. x^{9} 10. y^{6} 11. a^{12} 12. $9x^{2}$ 13. x^{3} 14. $x^{4}y^{6}$

○○○ Standard form

15. 4.5×10^{4} 16. 6.1×10^{-4} 17. 8×10^{-2}

○ Spot the error

18. negative power means one over: 1/9

○○○○○○○○ Coefficients and negatives

19. $3x^{3}$ 20. $4x^{6}$ 21. $3a^{4}b$ 22. −32 23. −16 24. −3 25. x^{3} 26. 16/9

○○○○ Calculating in standard form

27. 3×10^{9} 28. 3×10^{10} 29. 4.2×10^{3} 30. 2.5×10^{5}

○○○○○○ Find the unknown

31. n = −2 ($1/9 = 3^{-2}$) 32. x = 5 ($32 = 2^{5}$) 33. x = 3/2 ($2^{2x} = 2^{3}$) 34. x = 4 (x = 8^(2/3)) 35. k = 7 (k + 3 = 10) 36. x = 2 ($125 = 5^{3}$, x + 1 = 3)

10 Number bases

○○○ Binary to ordinary

1. 13 2. 32 3. 255

○○○ Ordinary to binary

4. 1001 5. 11001 6. 1100100

○○○○ Hexadecimal

7. 26 8. C8 9. pure blue 10. B7

○○○ Arithmetic

11. 1000_{2} (5 + 3 = 8) 12. 10000_{2} 13. $2^{4} = 16$

○○ Other bases

14. 17 15. octal has no digit 8

○○○○ More conversions

16. 144_8 (100 ÷ 8 = 12 r 4, 12 ÷ 8 = 1 r 4, 1 ÷ 8 = 0 r 1) 17. 1001_2 (14 − 5 = 9) 18. 111100_2 (0011 1100) 19. 153_8 (001 101 011)

○○○○○○ Find the unknown

20. x = 1 (8 + 2 + x = 11) 21. base 3 (2b + 1 = 7) 22. base 6 ($b^2 = 36$) 23. x = B (48 + x = 59, x = 11) 24. b = 6 (2b + 5 = 17) 25. x = 1 (4 + 2x + 1 = 7)

11 Simplifying

○○○ Collect like terms

1. 3a + 4b 2. $7x^2 + 4x - 2$ 3. $-2p - 2q$

○○○ Multiply

4. 12xy 5. $-10a^2$ 6. $6x^5$

○○○○ One bracket

7. 10x − 15 8. −3x + 12 9. $2x^2 + 7x$ 10. −x + 13

○○○○○ Two brackets

11. $x^2 + 7x + 10$ 12. $x^2 + 4x - 21$ 13. $2x^2 - 7x - 4$ 14. $x^2 - 12x + 36$ 15. $9x^2 - 4$

○○○ Minus signs outside

16. −2x + 7 17. $-3x^2 + 4x$ 18. −2x + 14

○○○○○○ Find the unknown

19. a = 4 (a + 3 = 7; check 3a = 12) 20. k = 4 (2k − 3 = 5) 21. p = 5 (2p = 10; p^2 = 25) 22. a = 4 (a + 3 = 7; −2a = −8) 23. b = 3 (constants: b = 3; check 2b + 1 = 7) 24. c = 4 (3c = 12)

12 Factorisation

○○○○ Common factor

1. 4(x + 3) 2. 3x(2x − 3) 3. 5a(b + 2) 4. $2x(x^2 + 2x + 4)$

○○○○○○○ Three terms: the pair adds to b and multiplies to ac (Δ shows whether it exists)

5. Δ = 1, pair 4, 3: (x + 3)(x + 4) 6. Δ = 81, pair 4, −5: (x − 5)(x + 4) 7. take out 3 first; Δ = 36, pair 4, −2: 3(x + 4)(x − 2) 8. Δ = 81, pair 10, 1: (2x + 1)(x + 5) 9. Δ = 1, pair 4, 3: (2x + 1)(3x + 2) 10. Δ = 64, pair 2, −6: (2x − 3)(2x + 1) 11. Δ = 289, pair 15, −2: (5x − 2)(x + 3)

○○○ Whole, square roots, or nothing? (Δ first)

12. $\Delta = 4 = 2^2$: pair 4, 2, (x + 2)(x + 4) 13. Δ = 8: pair $3 \pm \sqrt{2}$, $(x + 3 - \sqrt{2})(x + 3 + \sqrt{2})$ 14. Δ = −8: no brackets

○○○○ Difference of two squares

15. (x − 5)(x + 5) 16. (2x − 7)(2x + 7) 17. 2(x − 3)(x + 3) 18. $100^2 - 3^2 = 9991$

○○○ Grouping and chunks

19. (a + b)(x + y) 20. (x − 1)(x + 2) 21. x = b/(a − c)

○○ Cubes

22. $(x - 2)(x^2 + 2x + 4)$ 23. $(3x + 1)(9x^2 - 3x + 1)$

○○○○○○ Find the unknown

24. $p = 4$, $k = 7$ ($3p = 12$, $k = 3 + p$) 25. $k = 9$ ($(6 \div 2)^2 = 9$) 26. $q = 3$, $c = 6$ ($2 + q = 5$, $c = 2q$) 27. $k = -12$, $(x - 3)$ ($4 + p = 1$, so $p = -3$; $k = 4p$) 28. $k = 12$ ($(2x + 3)^2$, middle $2 \times 2x \times 3$) 29. $b = 5$ (expand: $6x - x = 5x$)

13 Algebraic fractions

○○ When is it undefined?

1. $x = 5$ 2. $x = 2$ or $x = -2$

○○○○○ Cancel (factorise first!)

3. $2x/3$ 4. $x + 2$ 5. $x - 1$ 6. $x + 3$ 7. can't cancel: it's $x/5 + 1$

○○○ Multiply and divide

8. $6/x$ 9. 5 10. x

○○○○○ Add and subtract

11. $3/x$ 12. $3/(2x)$ (LCD $2x$) 13. $(3y - 2x)/(xy)$ 14. $2x/((x + 1)(x - 1))$ 15. $(x + 5)/4$

○○○○○ Signs, nested fractions, equations

16. -1 17. $(x + 1)/(x + 3)$ 18. $x/(x - 2)$ ($\times x^2$, then cancel $x + 2$) 19. $x(x + 2)$ 20. $x = 3$ ($\times 3x$: $12 - x = 3x$)

○○○○○ Find the unknown

21. $A = 2$ ($3(x + 1) + A(x - 1) = 5x + 1$) 22. $k = 5$ ($(x + 2)(x + 3) = x^2 + 5x + 6$) 23. $k = 4$ ($\times x$: $1 + 1/k = 5/4$) 24. undefined at $x = 3$ (bottom 0), zero at $x = -2$ (top 0) 25. $a = 2$ ($ax - (x - 2) = x + 2$)

14 Transposing formulae

○○○○○ Fill in the chains

1. $x = (y + 4)/5$ 2. $V = PR$ 3. $a = (v - u)/t$ 4. $v = \sqrt{(2E/m)}$ 5. $A = 6s^2$

○○○○○ Minus signs, bottoms and cubes

6. $x = (7 - y)/3$ 7. $R = V/I$ 8. $x = 1/f - 3$ 9. $x = \sqrt[3]{(3V)}$ 10. $x = (3y + 1)/(y - 2)$

○○○○○○ Find the unknown

11. $s = 30$ ($s = (v^2 - u^2)/(2a) = 300/10$) 12. $t = 5$ ($t = (v - u)/a$) 13. $r = 7$ ($r^2 = A/\pi = 49$) 14. $C = 10$ ($C = 5(F - 32)/9$) 15. $x = 2$ ($x = (y + 1)/(y - 1)$) 16. $m = 5$ ($m = 2E/v^2 = 180/36$)

15 Solving equations

○○○○○ One unknown

1. $x = 5$ 2. $x = 4$ 3. $x = 6$ 4. $x = 14$ 5. 40 ml

○○○○○ Two unknowns

6. $x = 7$, $y = 3$ 7. $x = 2$, $y = 3$ 8. pen £1, pad £2 9. endless (same line) 10. about 25.5 ml and 44.5 ml

○○○○○○ Quadratics (Δ first)

11. ±7 12. Δ = 1: x = 2 or 3 13. Δ = 36: x = 2 or −4 14. x = 0 or 7 15. $(x + 4)^2 - 6$ 16. Δ = 12: −0.27 or −3.73

○○○○○○○ Methods, words and inequalities

17. x = 2, y = 5 18. x = 1, y = 2 19. $x^2 - x - 12 = 0$: x = 4 or −3 20. $3(x - 2)^2 - 7$ 21. x < 2 (open circle) 22. 6 m (reject −8) 23. x = 1 or 2

○○○○○○ Find the unknown

24. k = ±8 ($k^2 - 64 = 0$) 25. k = 1, other root −4 ($9 + 3k - 12 = 0$) 26. k > 9 ($36 - 4k < 0$) 27. k = 2, other root −1/2 ($8 - 10 + k = 0$) 28. y = 1, a = 3 ($3a + 2 = 11$) 29. k = 4 ($16 - 4k = 0$)

16 Matrices

○○○ Reading

1. 1 × 3 2. 4 × 2 3. 7

○○○ Adding and scaling

4. [[1, 7], [2, 6]] 5. [[6, −3], [0, 12]] 6. [[5, 2], [1, 0]]

○○○○ Multiplying

7. [14] 8. [[6], [12]] 9. [[5, 4], [1, 2]] 10. No: inner numbers 2 and 3 don't match

○○○ Inverse and equations

11. 4 − 6 = −2 12. [[3, −1], [−5, 2]] (determinant 1) 13. x = 2, y = 1

○○○○○○ Find the unknown

14. k = 3 ($4k - 12 = 0$) 15. k = ±3 ($k^2 - 9 = 0$) 16. a = 3, b = 4 ($2 + 3a = 11$, $b + 3 = 7$) 17. x = 3, y = −2 (match entries) 18. k = 2 (top-left of the product: $2k - 3 = 1$) 19. yes: the product is [[1, 0], [0, 1]] = I

17 Sequences and series

○○○○○ Arithmetic

1. 64 2. 20 3. 1275 4. 710 5. 3

○○ Sigma

6. 3 + 5 + 7 + 9 + 11 = 35 7. Σ from i = 1 to 10 of 4i (total 220)

○○○○○ Geometric

8. 4374 9. 1275 10. 24 11. 3/4 12. G.P. with r = 1.5; no (r > 1)

○○○○○○ Find the unknown

13. d = 4, a = 5 ($6d = 24$) 14. n = 12 ($3n^2 + 5n - 492 = 0$; reject the negative root) 15. r = 2, a = 3 ($r^3 = 96/12 = 8$) 16. r = 1/3 ($8/(1 - r) = 12$) 17. n = 9 ($S_8 = 3(2^8 - 1) = 765$, $S_9 = 1533$) 18. x = 3 ($(x + 3)^2 = x(x + 9)$, so $3x = 9$; terms 3, 6, 12)

18 Functions

○○○○ Using the notation

1. 5, −1, −13 2. 6, 12 3. 2, 20 4. 3a + 5

○○○ Domain and image

5. x = 4 6. x ≥ 1 7. [5, ∞): outputs of 5 or more

○○○○ Pieces, chains and inverses

8. £5, £10 9. 10, 18 10. $f^{-1}(x) = (x + 2)/5$ 11. $f^{-1}(x) = 4x - 1$

○○○○○○ Find the unknown

12. $x = 5$ ($5x = 25$) 13. $x = 2$ or $x = -6$ ($(x + 6)(x - 2) = 0$) 14. $a = 3$, $b = 1$ ($3a = 9$)
15. $x = 5$ ($fg(x) = 3(x + 2) - 1 = 3x + 5$) 16. $k = 3$ ($8k + 1 = 25$) 17. $a = 5$ ($f^{-1}(a) = 6$ means $a = f(6)$)

19 Graphs of functions

○○○ Points and plotting

1. 2nd (top-left) 2. $y = -5, -3, -1, 1, 3$ (a straight line) 3. y-int 0; $x = 0$ and $x = 4$

○○○○ Even, odd, neither

4. even 5. odd 6. neither (try $x = 1$ and $x = -1$: 2 and 0) 7. odd

○○○○ Recognise and move

8. negative powers (even: always positive) 9. (1, 5) 10. moved left 4 11. x^4

○○○○○○ Find the unknown

12. $h = 3$, $k = -2$ 13. (2, 1): inside +4 moves left 4 14. (1, 1): left 1, down 4 15. (4, 6): $y \times (-2)$ 16. $a = 2$ ($7 = 4a - 1$) 17. $k = -2$ (denominator $-2 - k = 0$)

20 The straight line

○○○○ Reading y = mx + c

1. $m = 5$, $c = -2$ 2. $m = -3$, $c = 7$ 3. $y = -2x + 5$: $m = -2$, $c = 5$ 4. $y = 30x + 40$

○○○○ Two points

5. 3 6. $y = 3x - 1$ 7. $y = -2x + 3$ 8. $y = 3x + 5$

○○○ Sketch and perpendicular

9. (0, 3) and (5, 0) 10. $-1/4$ 11. $3/2$

○○○○ Parallel, perpendicular or neither?

12. parallel 13. perpendicular 14. neither 15. perpendicular

○○○ Lines through a point

16. $y = 2x - 5$ 17. $y = -x/2 + 2$ 18. $y = -x/2 + 5$

○○○○ Midpoint, distance, intersection

19. (2, 6) 20. 5 21. (5, 7) 22. (3, 7)

○○○○○○ Find the unknown

23. $k = 5$ 24. $k = 4$ 25. $k = -1/4$ 26. $a = 2$ 27. yes 28. $y = -2x + 7$; (3.5, 0)

21 The exponential function

○○○○ Power or exponential?

1. power 2. exponential growth 3. exponential decay 4. power

○○○ Using $y = A \times b^x$

5. $200 \times 2^6 = 12{,}800$ 6. $15{,}000 \times 0.8^3 = £7680$ 7. $500 \times 1.03^4 \approx £562.75$

○○○ e

8. 7.39 and 0.14 9. about 62.5 °C 10. 4 A and $4e^{-1} \approx 1.47$ A

○○○○○○ Find the unknown

11. A = 5, b = 3 ($b^2 = 45/5 = 9$) 12. b = 2, A = 5 ($b^2 = 40/10 = 4$) 13. 12.5 mg (4 half-lives: $200 \times (1/2)^4$) 14. 80 minutes ($4800/300 = 16 = 2^4$) 15. 6% ($m^2 = 1.1236$, m = 1.06) 16. a 15% decrease (1 − 0.85)

22 Logarithms

○○○○○○ What is it?

1. 2 2. 5 3. −3 4. 0 5. $\log_5 125 = 3$ 6. $2^6 = 64$

○○○ Laws

7. log 20 8. log 2 (since 8/4 = 2) 9. 3 log x + log y − log z

○○○○○ Solve

10. 4 11. 2.29 12. ln 20 ≈ 3.00 13. log 2 / log 1.08 ≈ 9.0 years 14. $t = (1/\lambda) \ln(N_0/N)$

○○○○○○ Find the unknown

15. x = 64 ($x = 2^6$) 16. x = $e^3 \approx 20.09$ 17. x = 5 ($x^2 - 3x - 10 = 0$; reject x = −2) 18. x = 0.557 (2x + 1 = log 30 / log 5 = 2.113) 19. x = 1.756 (x = log 5 / (log 5 − log 2)) 20. k = 0.458 (e^(−2k) = 0.4, k = −ln 0.4 / 2)

23 Tables and charts

○○○○ Which chart?

1. pie 2. histogram 3. scatter plot 4. line graph

○○ Build it

5. 1 | 2 5; 2 | 1 2 2 7; 3 | 0 4; 4 | 1; key 2 | 1 means 21 6. 25/90 × 360 = 100°

○○○ Detective

7. correlation ≠ causation (keen students choose the front) 8. 99 looks 4× bigger than 96; start at 0 9. only if it's an error; check it, and say so if you remove it

○○○○○○ Find the unknown

10. 72/360 × 250 = 50 11. 45° is 1/8 of 360°, so 30 × 8 = 240 12. 3.5 × 10 = 35 (frequency = density × width) 13. 24 ÷ 3 = 8 (density = frequency ÷ width) 14. 2 × 10 + 6 × 5 + 1 × 15 = 20 + 30 + 15 = 65 15. x = 360 − 150 − 120 = 90°; 90/360 × 120 = 30

24 Statistics

○○○○ Averages

1. 62 ÷ 7 ≈ 8.86 2. 8 3. 7 4. the mean (it becomes 20.25; median 9)

○○○○ Spread

5. 12 6. 6th; (6 + 1)/2 = 3.5th from each end (halfway between the 3rd and 4th) 7. 8/3 ≈ 2.67, σ ≈ 1.63 8. √(8/2) = 2

○○ Think

9. a few mansions skew the mean upward 10. the spread (standard deviation / IQR)

○○○○○○ Find the unknown

11. 8 (total 5 × 8 = 40; 40 − 32 = 8) 12. x = 6 (6x = 3 × 12 = 36) 13. 17 (60 + 25 = 85; 85 ÷ 5) 14. 8 (60 − 20 = 40; 40 ÷ 5) 15. (a) mean 12, σ = 2 (b) mean 21, σ = 6 16. k = 4 (Σfx = 10 + 2k = 1.8 × (6 + k))

25 Probability and Bayes

○○○○○ Two dice

1. 6/36 = 1/6 2. 4/36 = 1/9 3. 5/6 4. (18 + 6 − 3)/36 = 21/36 = 7/12 5. 1/6 (only blue 5)

○○○ Cards and tables

6. 4/52 × 3/51 = 1/221 7. 15/30 = 0.5 8. yes; 1/2 × 1/6 = 1/12

○ Bayes

9. 20 ill, 18 positive; 980 well, 49 positive; 18/67 ≈ 0.27

○○○○○○ Find the unknown

10. 0.6 + 0.5 − 0.8 = 0.3; 0.6 × 0.5 = 0.3, so independent 11. 0.4 × 0.25 = 0.1 12. 0.12 ÷ 0.3 = 0.4 13. 0.8 − 0.35 = 0.45 (no overlap) 14. 0.7 = 0.4 + p − 0.4p, so p = 0.5 15. bus 120, late 30; others 180, late 18; 48 late; 30/48 = 0.625

26 The Normal distribution

○○○○ z-scores

1. 1 2. −2 3. −1 4. 100 + 1.645 × 15 ≈ 124.7

○○○○ Areas

5. Φ(2) = 0.977 6. 1 − Φ(1) = 0.159 7. 0.683 (the 68% rule) 8. 0.95 − 0.5 = 0.45

○○ Backwards

9. 60 + 1.28 × 10 ≈ 72.8 10. the average of 20: sums of many independent bits become bell-shaped

○○○○○○ Find the unknown

11. Φ(z) = 0.10, z = −1.2816; 100 − 1.2816 × 15 ≈ 80.8 12. z = ±1.96; 100 ± 29.4: 70.6 to 129.4 13. z = −1.645; μ = 250 + 1.645 × 4 ≈ 256.6 g 14. z = 1.2816; σ = (75 − 55)/1.2816 ≈ 15.6 15. 70 = μ + 2σ and 40 = μ − σ, so σ = 10, μ = 50 16. Φ(z) = 0.9332, z = 1.5; σ = 15/1.5 = 10

27 Angles

○○○○ Degrees

1. 120° 2. 105° (hour hand at 285°, minute at 180°) 3. obtuse 4. a quarter turn clockwise

○○○○ Converting

5. 2π/3 6. 30° 7. 143.2° 8. 5π/4

○○○ Arcs

9. 12 cm 10. 4π ≈ 12.6 m² 11. 6 ÷ 12 = 0.5 rad

○○○○○ Find the unknown

12. r = 6 cm (½r² × 1.5 = 27, so r² = 36) 13. θ = 15 ÷ 6 = 2.5 rad ≈ 143.2° 14. r = 27 cm ((40/360) × 2πr = 6π, so 2πr/9 = 6π) 15. 120° ((θ/360) × 81π = 27π, so θ/360 = 1/3) 16. 750° − 720° = 30°; −5π/3 + 2π = π/3

28 Intro to trigonometry

○○○ Pythagoras

1. 13 2. 8 3. √(4 + 9 + 36) = 7

○○○ Sides

4. 8 sin 40° ≈ 5.14 5. 6 tan 25° ≈ 2.80 6. 4 ÷ sin 55° ≈ 4.88

○○○ Angles

7. $\sin^{-1}(5/13) \approx 22.6°$ 8. $\cos^{-1}(0.7) \approx 45.6°$ 9. $\tan^{-1}(1/12) \approx 4.8°$

○○○○○ Find the unknown

10. sin θ = 5/6, θ ≈ 56.4° 11. 30 tan 28° ≈ 15.95 m 12. 45 ÷ tan 18° ≈ 138.50 m 13. sin 5° = 0.8/H, H = 0.8 ÷ sin 5° ≈ 9.18 m 14. O = 0.6 × 15 = 9; H = √(81 + 225) ≈ 17.49 15. 40 tan 31° − 40 tan 25° ≈ 24.03 − 18.65 = 5.38 m

29 Trig graphs

○○○ Circle

1. (0, 1) 2. negative (left half) 3. −0.5

○○○○ Graphs

4. 2π, 4 5. 2π/3 6. 4π 7. x = π/2 and 3π/2

○○ Wheels

8. h = 12 − 10 cos(2πt), t in minutes 9. 0.5 − 0.125/6 + 0.03125/120 ≈ 0.479427

○○○○ Recipes

10. 1 − 0.25/2 + 0.0625/24 ≈ 0.877604 (true 0.877583) 11. 0.3 + 0.027/3 = 0.309 (true 0.3093) 12. sin 0.866 < arc 1.047 < tan 1.732 13. the upright side, h cos θ: sin grows at cos's rate

○○○○○ Find the unknown

14. cos 225° = −cos 45° = −1/√2 (Q3); tan 300° = −tan 60° = −√3 (Q4) 15. α = $\sin^{-1}$ 0.6 = 36.87°, θ = 180° − 36.87° = 143.13°; cos θ = −0.8, tan θ = −0.75 16. D = (7 + 1)/2 = 4, A = (7 − 1)/2 = 3, B = 360/90 = 4 17. B = 360/72 = 5 18. period 45° = 180°/B, so B = 4

30 Trig identities and equations

○○○ Identities

1. 12/13 2. 3 3. no: sin 180° = 0 but 2 sin 90° = 2

○○○○ Equations

4. 53.13°, 126.87° 5. 60°, 300° 6. 45°, 225° 7. 210°, 330°

○○○ r sin form

8. 5 sin(θ + 53.13°) 9. 5 10. θ + 53.13 = 90, so θ = 36.87°

○○○○○ Find the unknown

11. $\theta_1 = -36.87°$; $180° - (-36.87°) = 216.87°$, $-36.87° + 360° = 323.13°$ 12. $\theta_1 = \cos^{-1}(-0.25) = 1.823$; $2\pi - 1.823 = 4.460$ 13. $\theta_1 = -56.31°$; $+ 180° = 123.69°$, $+ 360° = 303.69°$ 14. $2\theta = 60°, 300°, 420°, 660°$ (range 0°–720°), so $\theta = 30°, 150°, 210°, 330°$ 15. $\sin^2\theta = 1 - 25/169 = 144/169$, $\sin\theta = -12/13$ (Q3); $\tan\theta = 12/5$

31 Solving triangles

○○○ Sine rule

1. $8 \sin 75°/\sin 40° \approx 12.02$ 2. $\sin^{-1}(9 \sin 110°/12) \approx 44.8°$ 3. third angle 80°; $20 \sin 65°/\sin 80° \approx 18.41$

○○○ Cosine rule

4. ≈ 6.13 5. $\cos^{-1}((16 + 49 - 81)/56) \approx 106.6°$ 6. 120°: $\sqrt{(149 + 70)} \approx 14.80$ cm

○○ Choose

7. no, only their ratios (the triangle could be any size) 8. 13.5

○○○○○ Find the unknown

9. $\sin C = 48/80 = 0.6$, $C = 36.87°$ or $143.13°$ 10. $35 = ½ \times 10 \times b \times 0.5$, so $b = 14$ cm 11. $C = 70°$; $a = 15 \sin 40°/\sin 70° \approx 10.26$; area $= ½ \times 10.26 \times 15 \times \sin 70° \approx 72.31$ 12. $c^2 = 49 + 81 - 63 = 67$, $c \approx 8.19$; area $= ½ \times 7 \times 9 \times \sin 60° \approx 27.28$ 13. A is obtuse, so one triangle: $\sin B = 5 \sin 120°/9 \approx 0.481$, $B \approx 28.76°$

32 Vectors

○○○ Trips, adding, lengths

1. $2i + 3j$; $4i - 7j$ 2. $\sqrt{13} \approx 3.61$ 3. $-3i + 15j$; $\sqrt{234} \approx 15.30$

○○○ Dot product

4. $8 - 3 = 5$ 5. $\cos^{-1}(5 \div (\sqrt{5} \times \sqrt{10})) = 45°$ 6. yes: $12 - 12 = 0$

○○○ Cross product

7. $3\times2 - 1\times1 = 5$ 8. $6i - 3j + k$ 9. $6 - 6 + 0 = 0$ and $0 - 3 + 3 = 0$

○ Distance or displacement?

10. 7 km; $\sqrt{(9 + 16)} = 5$ km

○○○○○○ Find the unknown

11. $2k - 4 = 0$, $k = 2$ 12. $t = 2$ (from $3 \to 6$), so $k = 4$ 13. length 13: $(5/13)i - (12/13)j$ 14. $b - a = 6i - 8j$; $\sqrt{(36 + 64)} = 10$; $((1 + 7)/2, (3 - 5)/2) = (4, -1)$ 15. $15 \times (0.6i + 0.8j) = 9i + 12j$ 16. $\tan^{-1}(1) = 45°$; $x < 0$ so $+ 180°$: 225°

33 Polar coordinates

○○ Polar to Cartesian

1. (3, 5.20) 2. (−1.41, 1.41)

○○○○ Cartesian to polar (mind the quadrant!)

3. (5, 53.13°) 4. (5, 126.87°) 5. (2.83, 225°) 6. (7.07, 315°)

○○○○ Curves and equations

7. a circle of radius 3 8. $r = 1$, the point (0, 1) 9. $r \cos\theta = 4$, so $r = 4/\cos\theta$ 10. $r^2 = 25$, so $r = 5$

○ Double angle

11. $2 \times 0.7071 \times 0.7071 = 1$

○○○○○ Find the unknown

12. × r: $r^2 = 6r \sin\theta$, $x^2 + y^2 = 6y$, $x^2 + (y - 3)^2 = 9$: circle, centre (0, 3), radius 3 13. $\tan\theta = \sqrt{3}$, $\theta = \pi/3$ (60°) 14. on the negative y axis: (5, 270°) = (5, 3π/2) 15. $\sin\theta = 4/8 = 0.5$ and $x < 0$, so $\theta = 150°$; $x = 8\cos 150° \approx -6.93$ 16. $y = x\tan 135° = (-3)(-1) = 3$; $r = \sqrt{(9 + 9)} \approx 4.24$

34 Gradients of curves

○ Gradient from scratch

1. $(9.0601 - 9) \div 0.01 = 6.01$ (the exact answer is 6)

○○○○ Power rule

2. $20x^3$ 3. $6x - 7$ 4. ½x^(−½) = 1/(2√x) 5. $-12x^{-4} = -12/x^4$

○○○ Standard derivatives

6. 6e^(2x) 7. 10 cos 5x 8. 1/4 = 0.25

○○ Stationary points

9. $x = 3$, $y = -8$: minimum (second derivative $2 > 0$) 10. (1, −2) minimum ($6 > 0$); (−1, 2) maximum ($-6 < 0$)

○○○ Tangents and normals

11. point (1, 3), $m = 4$: $y = 4x - 1$ 12. $m = -1/4$: $y = -x/4 + 13/4$ 13. $dy/dx = -6/x^2 = -3/2$: $y = -3x/2 + 6$

○○○○○ Find the unknown

14. $a = 2$ ($2a + 3 = 7$) 15. (4, −2) ($2x - 5 = 3$, so $x = 4$) 16. $k = -12$ ($3(2)^2 + k = 0$) 17. (3, −8) and (−3, 10) ($3x^2 - 12 = 15$, so $x^2 = 9$) 18. $b = -3$ ($4 + b = 1$); point (2, −2), tangent $y = x - 4$

35 Chain, product, quotient

○○ Higher derivatives

1. $12x^2 - 6$ 2. $\sin x$

○○○ Chain rule

3. $10(2x + 1)^4$ 4. 6x e^(3x²) 5. $3x^2 \div (2\sqrt{(1 + x^3)})$

○○ Product rule

6. $2x\sin x + x^2\cos x$ 7. $e^x + x\,e^x = (1 + x)e^x$

○○ Quotient rule (or rewrite!)

8. $-2/(x - 1)^2$ 9. $1 - 1/x^2$ (easiest: $x + x^{-1}$ first)

○○ Implicit

10. $-x/y$; $-3/4$ 11. $1/(1 - 3y^2)$

○○○ Exam combinations

12. dy/dx = e^(−2x)(1 − 2x) = 0, $x = ½$; (½, 1/(2e)), maximum 13. $(x^2 + 2x)/(x + 1)^2 = 3/4$ 14. $2x/(x^2 + 1)$; 1

○○○○○ Find the unknown

15. a = 6 (a e^0 = 6) 16. a = 6 (4a(1)³ = 24) 17. k = 3 (2/(1 + k) = ½) 18. c = 4 (c/c² = 1/c = ¼) 19. b = 2 (4 + b³ = 12); dy/dx = −2x/(3y²) = −1/3

36 Integration and areas

○ Trapezium rule

1. 3 and 2.75

○○○○○ Indefinite integrals

2. 2x³ + c 3. x³ − 2x² + 5x + c 4. 2 ln|x| + c 5. ⅓e^(3x) + c 6. 2 sin 2x + c

○ Find the curve

7. y = 3x² + 2x + 5

○○○ Definite integrals

8. 9 9. 14 10. ½ (each side is ¼; the integral itself is 0)

○○○○○ Rewrite first

11. −2/x² + c (4x⁻³ → 4x⁻²/(−2)) 12. 4x^(3/2) + c 13. x³/3 + 2x² + 4x + c 14. x³/3 − 5 ln|x| + c 15. 2 (crosses at x = 1: pieces −⅔ and 4/3)

○○○○○ Find the unknown

16. y = 2x² + 3x − 3 (2 + 3 + c = 2) 17. k = 4 (k² − 1 = 15) 18. a = 2 (9a/2 + 6 = 15) 19. k = 4 (k³ = 64) 20. f(x) = 2x³ + 2/x + 1 (2 + 2 + c = 5)

37 Parts and substitution

○○○○ Integration by parts

1. x sin x + cos x + C 2. ½x e^(2x) − ¼e^(2x) + C 3. e^x(x² − 2x + 2) + C 4. 1

○○○○ Integration by substitution

5. (2x − 5)⁷/14 + C 6. (x² + 4)⁶/2 + C 7. e^(sin x) + C 8. 15/4 (limits become 1 and 2)

○○○ Which technique?

9. substitution: −½cos(x²) + C 10. parts: sin x − x cos x + C 11. table: −⅕cos 5x + C

○○○○ Linear inside shortcut

12. (5x − 2)⁴/20 + C 13. −⅓cos(3x + 1) + C 14. −½e^(−2x) + C 15. ¼ ln|4x + 1| + C

○○○○○ Find the unknown

16. k = 1 ([x e^x − e^x] gives (k − 1)e^k + 1 = 1) 17. a = 2 (1/(5a) = 1/10) 18. k = √3 (((k² + 1)⁴ − 1)/4 = 255/4, so k² + 1 = 4) 19. y = (x − 1)e^x + 4 (−1 + C = 3) 20. k = π/4 (½ sin 2k = ½, so 2k = π/2)

38 Partial differentiation

○○○ Freeze the other letter

1. 3x² + 4y, 4x + 2y 2. 10xy³, 15x²y² 3. ∂z/∂x = −3, ∂z/∂y = 7

○○ With the chain rule

4. 3cos(3x + 2y), 2cos(3x + 2y) 5. y e^(xy), x e^(xy)

○○ The hill h = 12 − x² − 2y²

6. ∂h/∂x = −4, ∂h/∂y = 4 7. equally steep (both size 4); east goes down, north goes up

○ Second partials

8. both are 6xy²

○ Small changes

9. δA ≈ 5(0.1) + 10(0.1) = 1.5 (true: 1.51)

○○○ Product rule and notation

10. (1 + xy)e^(xy), x²e^(xy) 11. 2x sin(xy) + x²y cos(xy) 12. 3x²y², 2x³y, 6x²y

○○○○○ Find the unknown

13. a = 3 (2axy = 4a = 12) 14. b = −1 (3 + 3b = 0) 15. k = 2 (ky e⁰ = 3k = 6) 16. (3, −1) (2x − 6 = 0, 2y + 2 = 0) 17. h = 1 (2πrh = πr², so 4πh = 4π)

39 Exam pocket cards

Worked answers, line by line, are on the last pages of this chapter itself.

Slide Errata

Small slips found on the lecture slides while writing this book. Each one is explained where it comes up; here they are all together so you can correct your own copies. Grey with a red wavy line: what the slide says. Green: what it should say.

L01 **Combining fractions: subtraction** see Chapter 2

slide: $\frac{a}{b} - \frac{c}{d} = \frac{ad + bc}{bd}$ ✗

fix: $\frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd}$

L01 **Significant figures table, 4 s.f.** see Chapter 3

slide: $0.123456789 \to 0.1234$ ✗

fix: 0.1235 (rounded; 0.1234 is truncated)

L02 **Big example: (x+1)^3** see Chapter 11

slide: $(x+1)^3 = x^3 + 4x^2 + 3x + 1$ ✗

fix: $(x+1)^3 = x^3 + 3x^2 + 3x + 1$

L02 **Difference of two squares example** see Chapter 12

slide: Expand: $x^2 - 4y^2$ ✗

fix: Factorise: $x^2 - 4y^2$

L02 **MASSIVE example** see Chapter 11

slide: $\cdots + 21x^2 + \cdots = x^4 + 10x^3 + 44x^2 + 50x + 24$ ✗

fix: $\cdots + 12x^2 + \cdots = x^4 + 10x^3 + 35x^2 + 50x + 24$

L02 **Tricky factorising example** see Chapter 12

slide: $x^2 + 2y^2 + 2x^3 + 4xy = (x^2 + 2y)(x^2 + 2y + 2x)$ ✗

fix: $x^2 + 2y + 2x^3 + 4xy = (x^2 + 2y)(1 + 2x)$

L03 **Compounds 2, step (4) solved** see Chapter 15

slide: $y = \dfrac{20}{0.55} = 364$ ✗

fix: $y = \dfrac{20}{0.055} \approx 364$

L06 **Example 3, Pythagoras step** see Chapter 28

slide: $5^2 - 2.5^2 = a = 4.33$ ✗

fix: $a = \sqrt{5^2 - 2.5^2} = \sqrt{18.75} \approx 4.33$

L07 **Even powers list** see Chapter 9

slide: $x \times \cdots \times x \text{ (8 times)} = x^6$ ✗

fix: $= x^8$

L07 **Laws of logs 2, example for log a^n** see Chapter 22

slide: $\log a^2 + \log b^3 = \log a^2 b^3$ ✗

fix: e.g. $\log a^3 = 3 \log a$

L07 **Standard form definition** see Chapter 9

slide: $a \times 10^n,\ 1 \leq a \leq 10$ ✗

fix: $a \times 10^n,\ 1 \leq a < 10$

L10 **Determinant answer** see Chapter 32

slide: $\mathbf{a} \times \mathbf{b} = -55\mathbf{i} + 22\mathbf{j} - 11\mathbf{k}$ ✗

fix: $\mathbf{a} \times \mathbf{b} = -55\mathbf{i} + 22\mathbf{j} + 11\mathbf{k}$

L10 **Dot product 3: angle form example** see Chapter 32

slide: $|A| = 9.22,\ |B| = 16.12,\ \theta = 60° \Rightarrow C = 50$ ✗

fix: $9.22 \times 16.12 \cos 60° = 74.3;\ \text{for } C = 50,\ \theta \approx 70.3°$

L10 **Finding theta (cross product magnitude)** see Chapter 32

slide: $14.87 \sin 19.7° = 11.18$ ✗

fix: $14.87 \sin 19.7° \approx 5.0$

L11 **Polar angles diagram** see Chapter 27

slide: $225° = \frac{4\pi}{3}$ ✗

fix: $225° = \frac{5\pi}{4}$

L11 **Polar to Cartesian** see Chapter 33

slide: $y = x \sin\theta$ ✗

fix: $y = r \sin\theta$

L12 **Multiplying two matrices, bottom-left entry** see Chapter 16

slide: $3 \times 5 + 4 \times 8 = 47$ ✗

fix: $3 \times 5 + 4 \times 7 = 43$

L12 **Reversing the process** see Chapter 16

slide: Now let's calculate CD ✗

fix: Now let's calculate DC

L13 **GP sum formula** see Chapter 17

slide: $\sum_{i=1}^{n} a\frac{1-r^n}{1-r}$ ✗

fix: $S_n = a\frac{1-r^n}{1-r}$ (no $\sum$ needed)

L13 **GP sum to infinity** see Chapter 17

slide: $S_\infty = \frac{a}{r-1}$ ✗

fix: $S_\infty = \frac{a}{1-r}$

L13 **Geometric progression example (footnote)** see Chapter 17

slide: $3072 = 3 \times 2^{11}$ ✗

fix: $3072 = 3 \times 2^{10}$

L13 **Mapping: the squaring function** see Chapter 6

slide: $\mathbb{R} \to \mathbb{Q}^+$ ✗

fix: $\mathbb{R} \to [0, \infty)$ (non-negative reals)

L13 **Sets: {1,2,3,4} in set-builder form** see Chapter 6

slide: $\{x \mid 1 \leq x \leq 4\}$ ✗

fix: $\{x \mid 1 \leq x \leq 4,\ x \in \mathbb{Z}\}$

L14 **Finding stationary points** see Chapter 34

slide: differentiating finds the zeros ✗

fix: it finds where the gradient is zero (stationary points)

L15 **Combined example, multiplied out** see Chapter 35

slide: $340x^4 + 1728x^3 + \ldots$ ✗

fix: $320x^4 + 1728x^3 + \ldots$

L15 **Double differentiation example** see Chapter 35

slide: $\frac{d^2y}{dx^2} = 24x - 4 + \frac{1}{x^3}$ ✗

fix: $\frac{d^2y}{dx^2} = 24x - 4 + \frac{2}{x^3}$

L16 **Integral of 1/x** see Chapter 36

slide: $\int \frac{1}{x}\,dx = \ln x + C$ ✗

fix: $\int \frac{1}{x}\,dx = \ln|x| + C$

L16 **Limits: definite integral of ax^n** see Chapter 36

slide: $\frac{a}{n+1}\left(b^{n+1} - c^{n+1}\right)$ ✗

fix: $\frac{a}{n+1}\left(c^{n+1} - b^{n+1}\right)$ (top limit minus bottom)

L16 **Trapezium rule** see Chapter 36

slide: $\frac{h}{2}(y_0 + y_1 + \cdots + y_n)$ ✗

fix: $\frac{h}{2}\big(y_0 + 2(y_1 + \cdots + y_{n-1}) + y_n\big)$

L17 **Definite integral example (parts)** see Chapter 37

slide: $\big[-xe^{x}\big]_0^2, \quad (-2e^2 - 0) - (e^2 - (-1))$ ✗

fix: $\big[-xe^{-x}\big]_0^2, \quad (-2e^{-2} - 0) - (e^{-2} - 1) = 1 - 3e^{-2}$

L17 **Substitution, second example** see Chapter 37

slide: $\int \frac{\sin y}{4}\,dy = \frac{\cos y}{4} + c$ ✗

fix: $\int \frac{\sin y}{4}\,dy = -\frac{\cos y}{4} + c = -\frac{\cos 2x^2}{4} + c$